

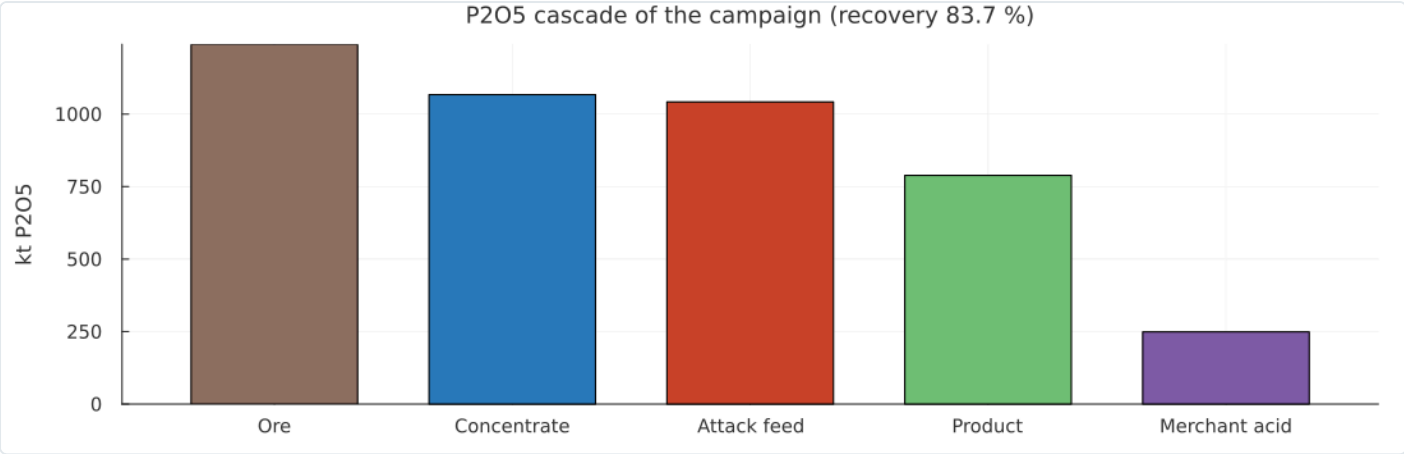
Atlas Phosphate Complex -- phosphate complex report

The ore, the acid, the product, the twin, the controller and the evidence

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1713.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 69 signals generated 2026-09-23T15:43:11.110 basis :screening_plant_model

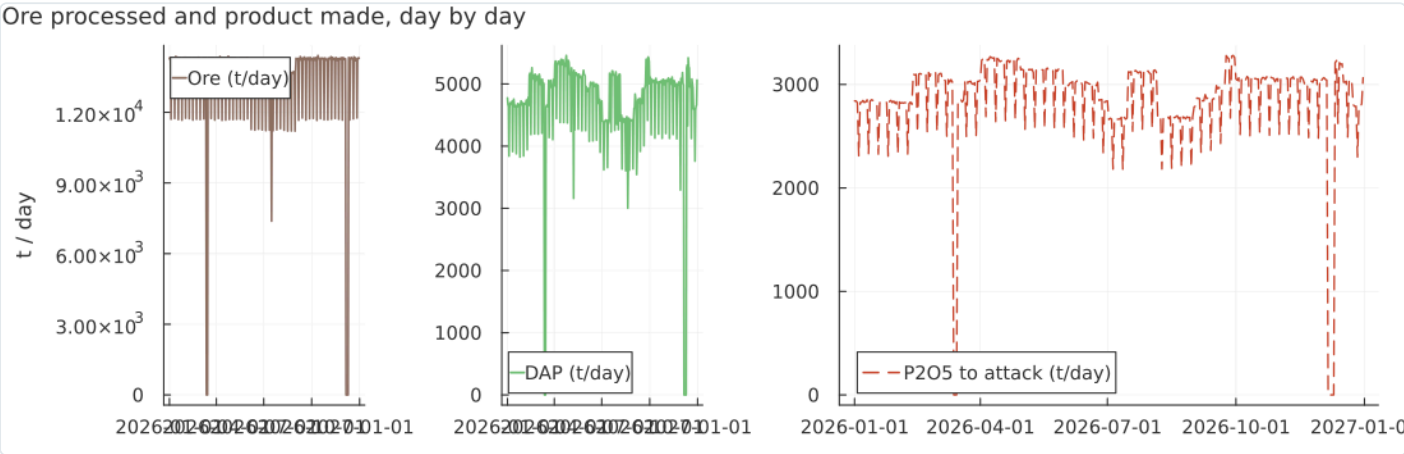
Executive summary

ORE PROCESSED 4910.27 kt 25.268 % P2O5 feed
CONCENTRATE 3326.974 kt 32.074 % P2O5, recovery 86.0 %
P2O5 TO ATTACK 1042.052 kt acid 2.823 t/t
ACID STRENGTH 55.852 % P2O5 steam 0.402 t/t
PRODUCT 1712.932 kt 18.142 % N, 88.5 % WSP
P2O5 RECOVERY 83.651 % plant, mine to bag
SPECIFIC ENERGY 2.007 GJ/t P2O5 56.8 kWh/t
VARIABLE COST 392.196 USD/t margin 651203.819 USD/t
CARBON 55.98 kgCO2e/t scope 1+2, 60342.0 t total
HISTORIAN 99.81 % usable 139 defect runs



The P2O5 cascade of the campaign, from the ore to the bagged product

Headline The complex processed **4,910.3 kt** of ore at 25.3 % P2O5 and delivered **1,712.9 kt** of Dap at 18.1 % N, recovering **83.7 %** of the P2O5 of the ore. The acid train digested 1,042.1 kt of P2O5 with 2.82 t of sulphuric acid per tonne, the filters lost 2.32 % of it in the gypsum and the evaporators spent 0.402 t of steam per tonne. The tonne of product cost **392.2 :USD** in variable cost and carried 55.98 kgCO2e per tonne of P2O5. The detector raised 100 findings worth 1.67734421e8 :USD a year, and the plant is **Noncompliant** against 42 clauses of 9 frameworks.



Ore processed and product made, day by day, over the campaign

The plant: flowsheet, design and instrumentation

REGISTERED DESIGN

Quantity	Value
Site	Atlas Phosphate Complex (Morocco)
Product	Dap
Design capacity	1,200 kt P2O5 per year over 8,000 h
Design rate	150 t P2O5/h, 601.44 t/h of ore
Registered units	34 unit operations in 9 areas, 57 streams
Instrument register	69 tags, 9 of them around the attack tanks
Ore bodies	boucraa, florida_pebble, gafsa, jacobina, khouribga, tapira
Revision	A -- capacities and targets are representative screening values

Unit	Area	Equipment	Kind	Feeds	Products	Capacity
:CRUSHING	:beneficiation	:jaw_crusher	:crush	:rom_ore	:crushed_rock	448
:GRINDING	:beneficiation	:rod_mill	:grind	:crushed_rock, :dilution_water	:mill_discharge	320
:CYCLONE_PACK	:beneficiation	:hydrocyclone_pack	:classify	:mill_discharge	:cyclone_overflow, :cyclone_underflow	320
:DESLIMING	:beneficiation	:desliming_cyclone	:deslime	:cyclone_overflow	:flotation_feed, :deslimed_slime	320
:ROUGHER_FLOTATION	:flotation	:rougher_cells	:flotation	:flotation_feed, :reagent_fatty_acid, :reagent_silicate	:rougher_concentrate, :scavenger_tailings	320
:CLEANER_FLOTATION	:flotation	:cleaner_cells	:flotation	:rougher_concentrate, :reagent_silicate	:cleaner_concentrate, :scavenger_tailings	160
:SCAVENGER_FLOTATION	:flotation	:scavenger_cells	:flotation	:scavenger_tailings, :reagent_fatty_acid	:final_concentrate, :flotation_tailings	192
:CONCENTRATE_THICKENING	:flotation	:concentrate_thickener	:thicken	:cleaner_concentrate, :final_concentrate	:thickener_overflow, :concentrate_cake	144
:CONCENTRATE_FILTRATION	:filtration	:concentrate_filter	:filter_dry	:concentrate_cake, :process_water	:dry_rock, :thickener_overflow	134.4
:FLUORINE_SCRUBBING	:filtration	:fluorine_scrubber	:scrub	:flash_vapour, :wash_water	:fluosilicic_acid, :stack_gas	150
:ATTACK_TANKS	:attack	:attack_tank	:digest	:dry_rock, :strong_sulphuric, :recycle_acid, :dilution_water	:attack_slurry	150
:DIGESTER	:attack	:digester	:digest	:attack_slurry	:digester_slurry	150
:FLASH_COOLING	:attack	:flash_cooler	:flash	:digester_slurry, :cooling_water	:filter_feed, :flash_vapour	150
:FILTRATION	:filtration	:belt_filter	:filter	:filter_feed, :wash_water	:filter_weak_acid, :gypsum_cake	150
:GYPSUM_DEWATERING	:filtration	:gypsum_belt	:dewater	:gypsum_cake, :wash_water	:process_water	150
:SULPHUR_MELTING	:acid_plant	:sulphur_melter	:melt		:molten_sulphur	150
:SULPHUR_BURNING	:acid_plant	:sulphur_burner	:burn	:molten_sulphur, :burner_air, :condensate	:so2_gas, :steam	345.48
:SO2_CONVERSION	:acid_plant	:converter	:convert	:so2_gas, :cooling_water	:so3_gas	345.48
:H2SO4_ABSORPTION	:acid_plant	:absorption_tower	:absorb	:so3_gas, :absorption_water	:strong_sulphuric	345.48
:EVAPORATION	:evaporation	:forced_circulation_evaporator	:evaporate	:filter_weak_acid, :steam	:concentrated_acid, :evaporator_condensate	150
:ACID_STORAGE	:evaporation	:acid_storage_tank	:store	:concentrated_acid	:recycle_acid	150
:SURFACE_CONDENSING	:evaporation	:surface_condenser	:condense	:evaporator_condensate, :cooling_water	:process_water	150
:AMMONIATION	:granulation	:ammonia_sparger	:ammoniate	:concentrated_acid, :ammonia	:dap_melt, :granulator_offgas	150
:GRANULATION	:granulation	:rotary_granulator	:granulate	:dap_melt, :undersize_recycle, :oversize_crush, :process_water	:green_pellets, :granulator_offgas	268
:DRYING	:granulation	:dryer_drum	:dry	:green_pellets, :natural_gas	:dried_pellets, :dryer_offgas	268
:COOLING	:granulation	:cooler_drum	:cool	:dried_pellets, :compressed_air	:cooled_pellets, :cooler_offgas	268
:SIZING	:granulation	:size_screens	:screen	:cooled_pellets	:product_dap, :oversize_crush, :undersize_recycle	268
:OVERSIZE_CRUSHING	:granulation	:crusher_oversize	:crush	:oversize_crush	:undersize_recycle	48.24
:COATING	:granulation	:coating_drum	:coat	:product_dap	:coated_product	268
:PRODUCT_SCRUBBING	:granulation	:product_scrubber	:scrub	:granulator_offgas, :dryer_offgas, :cooler_offgas, :process_water	:scrubber_liquor, :stack_gas	268
:STEAM_TURBINE	:utilities	:steam_turbine	:power	:steam	:electricity, :condensate	4.2
:COOLING_WATER	:utilities	:cooling_tower	:cool	:cooling_water_return	:cooling_water	9,800
:TAILINGS_THICKENING	:tailings	:tailings_thickener	:thicken	:flotation_tailings, :deslimed_slime	:thickener_overflow	192
:WATER_RECOVERY	:tailings	:process_water_pump	:recover	:thickener_overflow	:process_water	150

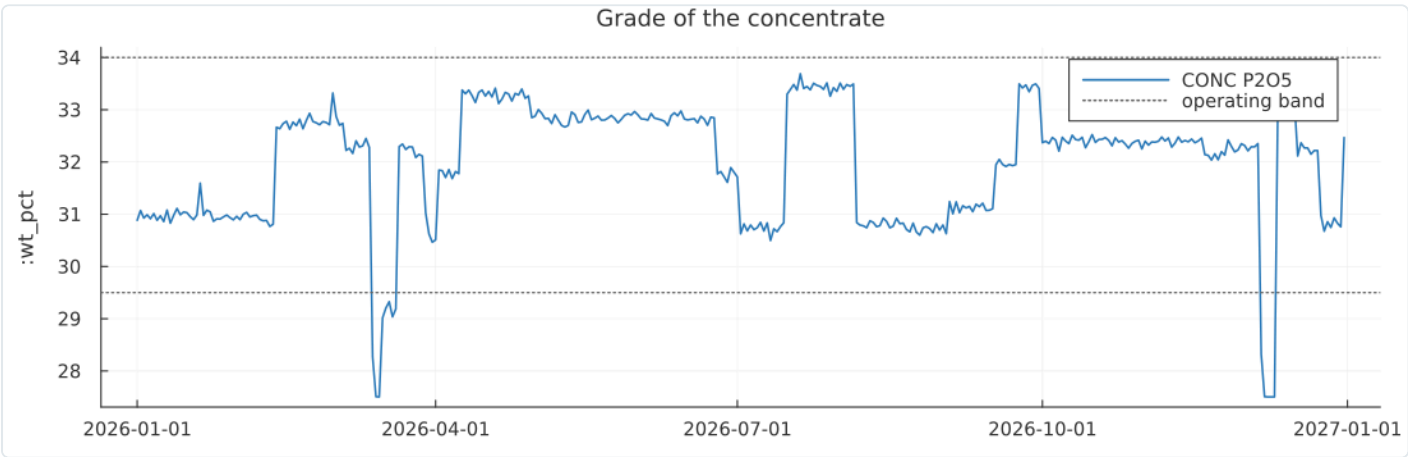
Ore preparation, flotation and the metallurgical balance

ASSAYS OF THE BLENDED ORE BODIES

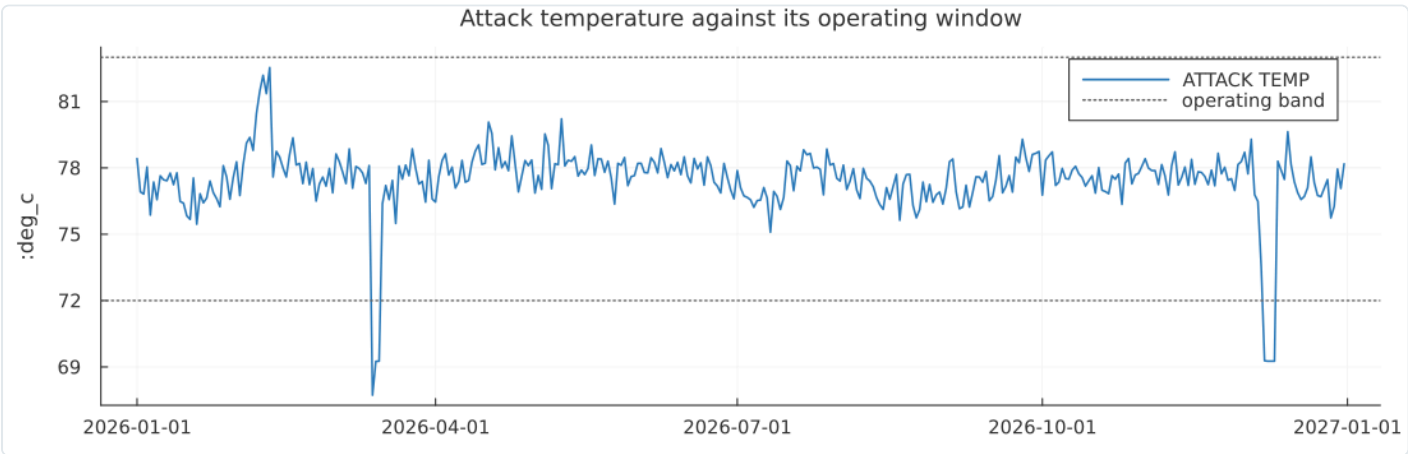
Ore body	P2O5 %	BPL %	CaO %	SiO2 %	Fe2O3 %	Al2O3 %	MgO %	F %	CO2 %	H2O %	CaO:P2O5	Gangue %
:boucraa	34.8	76.04	51	2.1	0.3	0.3	0.4	3.6	4.5	1	1.47	4.6
:florida_pebble	29.8	65.12	45.5	6	1.3	1.2	0.5	3.3	6.5	2	1.53	12.5
:gafsa	29	63.37	47	5.5	0.8	0.9	1.1	3.5	7.5	2.8	1.62	10.8
:jacobina	29.8	65.12	46	5	4.5	2.5	1.5	1.8	5.5	2.5	1.54	16.5
:khouribga	30.9	67.52	50.5	3	0.4	0.4	0.6	3.4	6	1.2	1.63	6.4
:tapira	33.4	72.99	48	4	3	2	1	2	4	1.5	1.44	12

METALLURGICAL BALANCE OF THE CAMPAIGN

Quantity	Value
Ore processed	4,910.3 kt
Flotation feed grade	25.27 % P2O5
Concentrate	3,327 kt at 32.07 % P2O5
Flotation recovery	86.01 %
Tailings grade	10.72 % P2O5
Grind P80	150.6 um
Specific grinding energy	11.39 kWh per t of ore
Reagent	0.9433 kg per t of ore
Two-product balance residual	-0.30898 %



P2O5 of the concentrate against the specification of the attack section

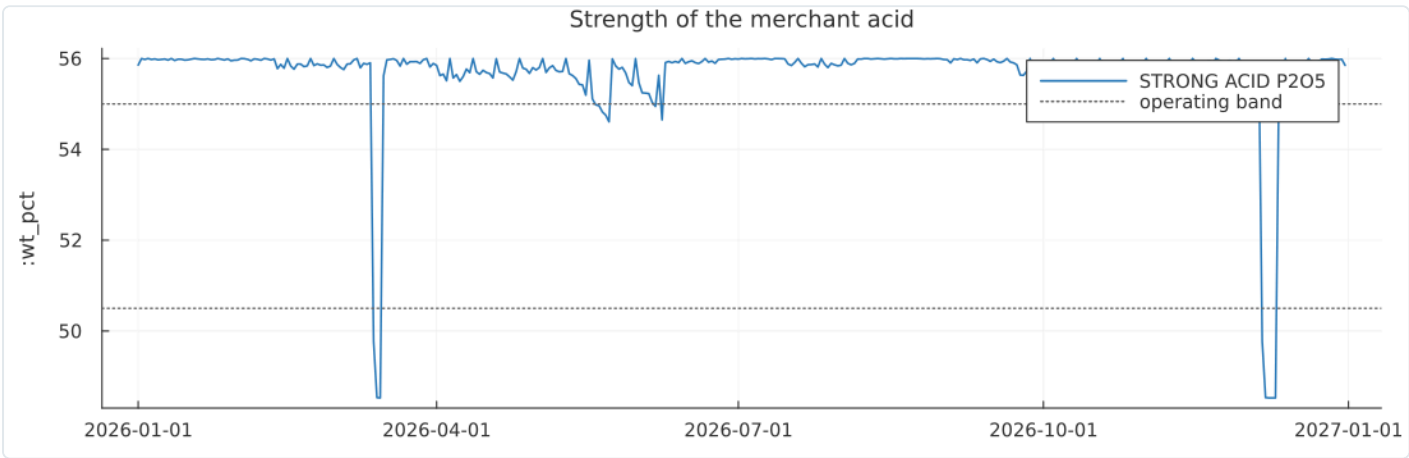


Temperature of the attack tanks, with the crystallisation window of the dihydrate process

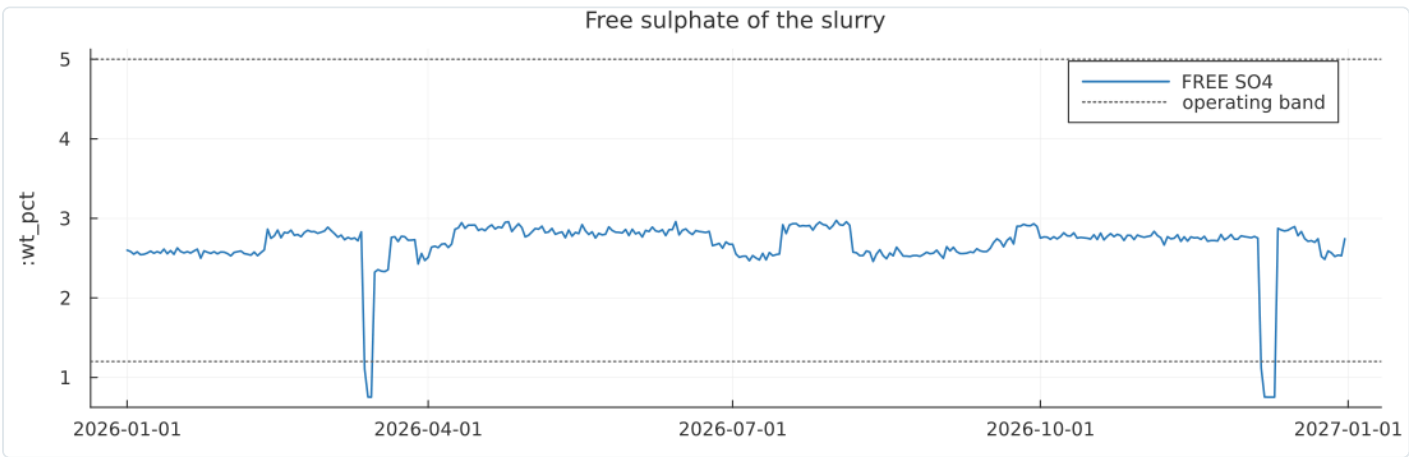
The reaction train: attack, filtration and evaporation

REACTION, FILTRATION AND EVAPORATION

Quantity	Value
P2O5 to the attack tanks	1,042.1 kt
Dry rock fed	3,277.1 kt
Sulphuric acid	2,942.2 kt (2.82 t per t P2O5)
Sulphur burnt	961.5 kt (0.9227 t per t P2O5)
Attack temperature	77.7 C
Free sulphate	2.72 wt% of slurry
Gypsum produced	4,846.3 kt (4.65 t per t P2O5)
Filter productivity	8.44 t P2O5 per m2 per day over 320 m2
Water-soluble loss	24.16 kt of P2O5 (2.32 % of the feed) at 0.49855 % free P2O5 in the cake
Filtrate strength	28.84 % P2O5
Evaporator steam	418.9 kt (0.402 t per t P2O5, economy 4.17)
Merchant acid	446 kt at 55.85 % P2O5



P2O5 of the merchant acid against the 52 % specification



Free sulphuric acid of the slurry, the crystallisation window of the gypsum

The acidulation demand of the blend is 2.82 t of H2SO4 per t of P2O5 against a stoichiometric 2.3 plus the registered excess of 2.5 %; the gap is the free carbonate of the ore, which is why the assay of the blend is an operating variable and not a laboratory curiosity.

Phosphoric acid: balance, specification and cost

The acid plant produced 446 kt of merchant acid -- 249.1 kt P2O5, 343.9 kt H3PO4 on the 100 % basis -- at 55.85 % P2O5, which is the :technical grade. The evaporators removed 1,574.8 thousand t of water with 0.41149 t of steam per t of P2O5 concentrated, an economy of 3.76 t of water per t of steam. The acid leaves the plant Compliant against the specification of the traded grade.

THE MERCHANT ACID OF THE CAMPAIGN

Quantity	Value
Merchant acid	446 kt (249.1 kt P2O5, 343.9 kt as H3PO4)
Grade	55.85 % P2O5 (77.12 % H3PO4) -- :technical
P2O5 of the attack	the filters receive 24.5 % of it as merchant acid, the rest goes to the granulation line
Yield	the filtration and the attack keep 97.68 % of the P2O5 fed; the evaporation closes at 101.96 %
Concentration	2.82 m3 of feed and 1.14 GJ of steam per t P2O5, 61 kWh of blower and pumps
Cost	503 :USD per t P2O5 as acid (281.1 per t of acid) against a price of 905 and a margin of 402 per t P2O5

THE MERCHANT ACID AGAINST THE SPECIFICATION OF THE TRADED GRADE

Specification	Instrument	Campaign	Limit	Unit	Direction	Margin %	Status
Colour	:ACID_COLOUR		2.86	4 :count	:le	28.5	Compliant
F	:ACID_F		0.3533	0.5 :wt_pct	:le	29.35	Compliant
Fe al	:ACID_FE_AL		0.1918	1 :wt_pct	:le	80.82	Compliant
P2o5	:STRONG_ACID_P2O5		55.85	52 :wt_pct	:ge	7.41	Compliant
So4	:ACID_SO4		1.16	1.5 :wt_pct	:le	22.39	Compliant
Solids	:ACID_SOLIDS		0.2943	0.5 :wt_pct	:le	41.13	Compliant

WHERE THE P2O5 OF THE ATTACK GOES

Destination	t P2O5	Share %	Note
Merchant acid	249,077	23.9	to the acid storage, the grade the customer buys
Fertiliser route	768,813	73.78	to the granulation line as concentrated acid
Gypsum cake	24,161	2.32	water-soluble P2O5 that left with the phosphogypsum
Unaccounted	0	0	sampling and rounding of the balance, targets zero

THE STEPS OF THE ACID ROUTE

Step	In (t P2O5)	Out (t P2O5)	Loss (t P2O5)	Yield %	Note
Attack and filtration	1.042052e6	1.017891e6	24,161	97.68	the cake carries the free P2O5 out of the circuit
Concentration	1.017891e6	244,294	0	101.96	the closure of the evaporation against the registered merchant split
Merchant acid	244,294	249,077	0	101.96	the 24.5 % of the filtered P2O5 the storage tank receives

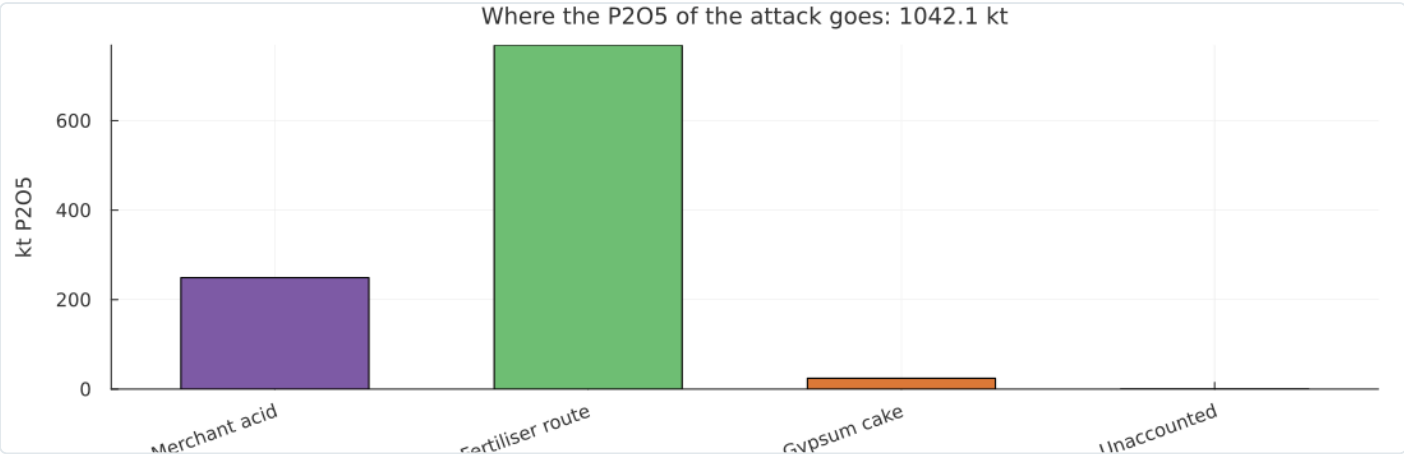
WHAT A TONNE OF P2O5 COSTS IN ROCK, ACID, STEAM AND WATER

Consumption	Value	Unit	Basis	Note
Rock	3.22	:t_ph	:per_p2o5_sold	t of rock per t P2O5 sold as acid, allocated by the acid share
Sulphuric acid	2.89	:t_ph	:per_p2o5_sold	t of H2SO4 per t P2O5 sold as acid
Sulphur	0.9446	:t_ph	:per_p2o5_sold	t of sulphur burnt per t P2O5 sold as acid
Steam	0.4115	:t_ph	:per_p2o5_concentrated	t of steam per t P2O5 the evaporators concentrate, both products included
Electricity	60.82	:kwh_per_t	:per_p2o5_concentrated	kWh per t P2O5 concentrated: blower of the acid plant and evaporator set
Water	2.82	:m3_ph	:per_p2o5_concentrated	m3 of evaporator feed per t P2O5 concentrated
Thermal energy	1.14	:gj_per_t	:per_p2o5_concentrated	GJ of steam per t P2O5 concentrated
Gypsum	4.76	:t_ph	:per_p2o5_sold	t of phosphogypsum per t P2O5

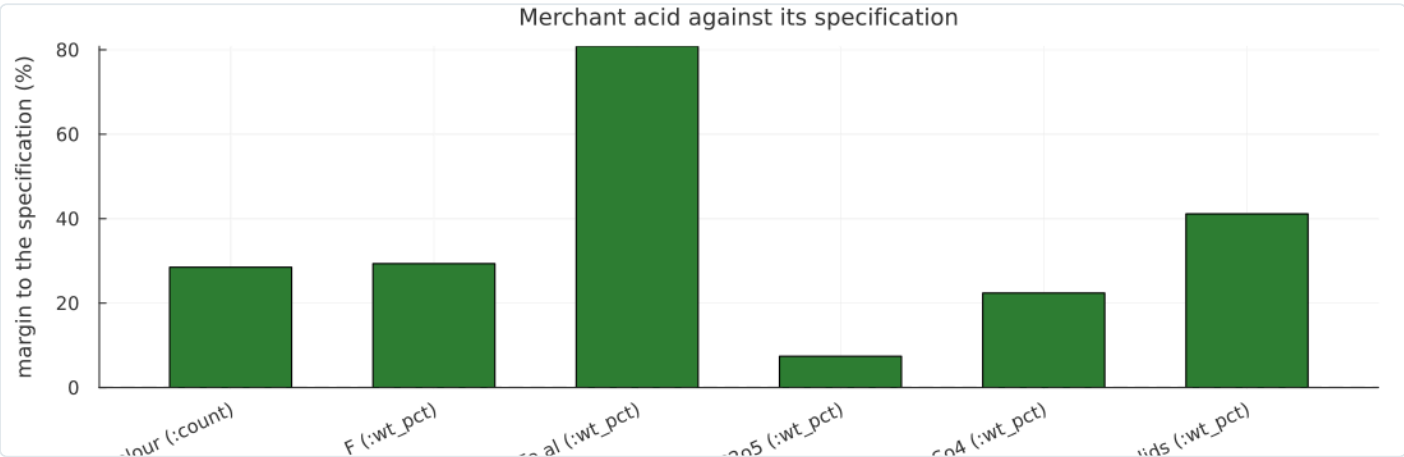
THE TARGETS OF THE ACID ROUTE

Metric	Campaign	Target	Unit	Direction	Margin %	Status
:acid_concentration_yield		101.96	97 :pct	:ge	5.11	Compliant
:acid_solids		0.2943	0.5 :wt_pct	:le	41.13	Compliant
:acid_electricity		60.82	75 :kwh_per_t	:le	18.9	Compliant
:acid_so4		1.16	1.5 :wt_pct	:le	22.39	Compliant
:acid_steam		0.4115	0.62 :t_ph	:le	33.63	Compliant

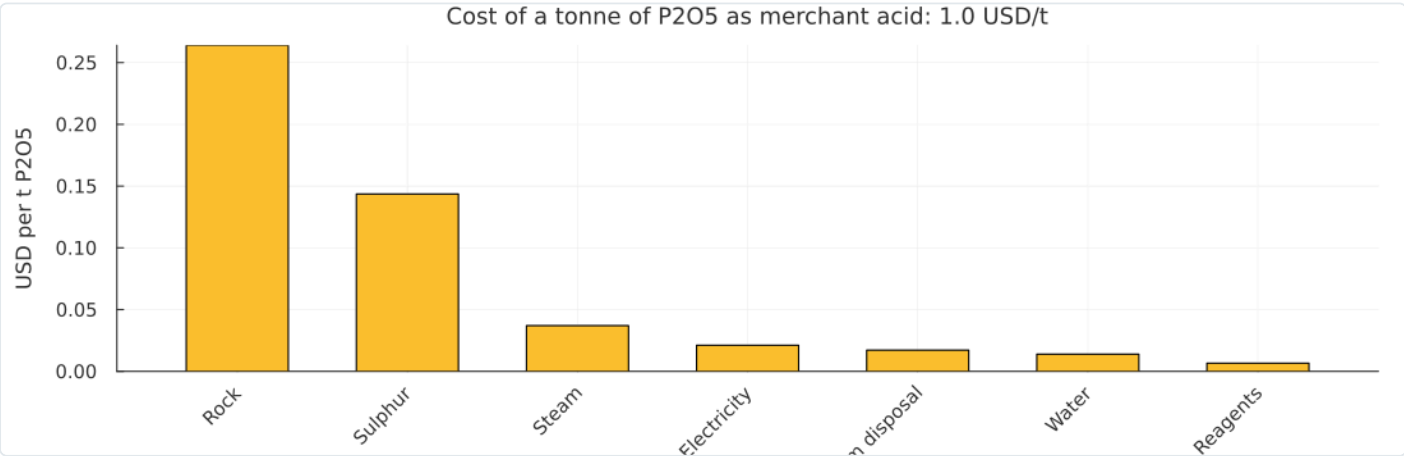
Metric	Campaign	Target	Unit	Direction	Margin %	Status
acid_cost		503.33	600 :usd_per_t	:le	16.11	Compliant
acid_f		0.3533	0.5 :wt_pct	:le	29.35	Compliant
acid_p2o5_yield		97.68	96.5 :pct	:ge	1.22	Compliant
acid_strength		55.85	52 :wt_pct	:ge	7.41	Compliant



Where the P2O5 of the attack goes: the acid route and the fertiliser route



The merchant acid against the specification of the traded grade



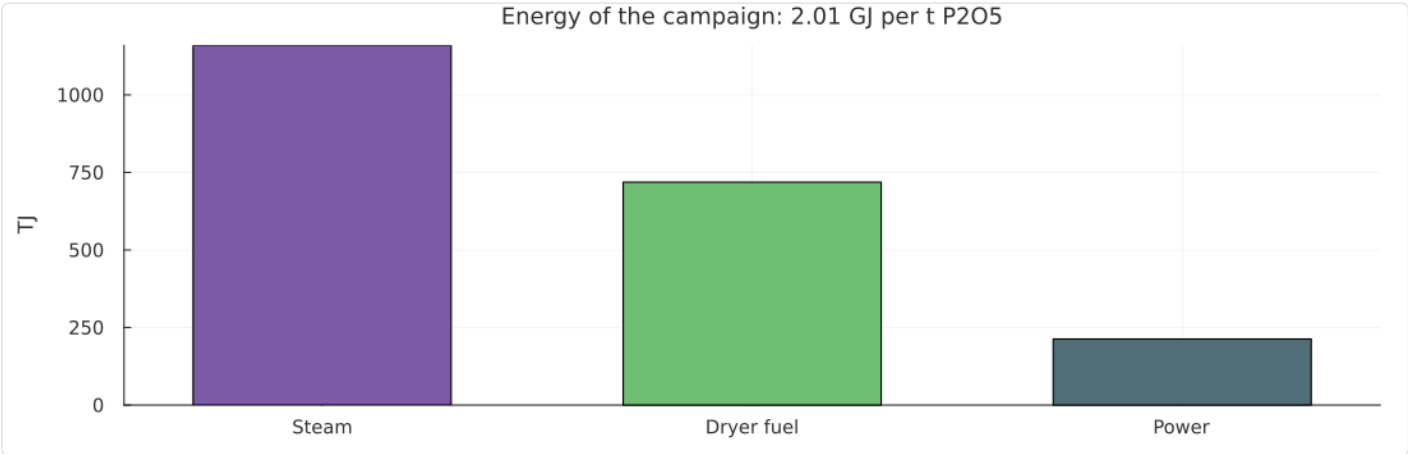
Cost of a tonne of P2O5 as merchant acid, item by item

Acid route: the plant kept 97.68 % of the P2O5 at the filters, sold 446 kt of acid at 55.85 % P2O5 for 905 per t P2O5, and spent 503.

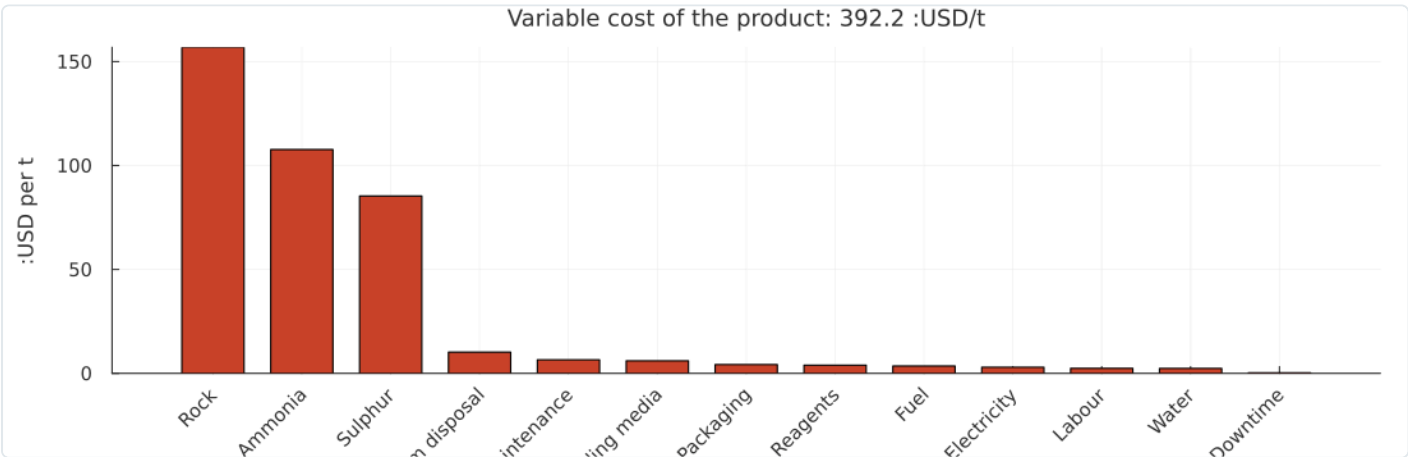
Granulation, product quality, energy, cost and carbon

GRANULATION, PRODUCT AND THE ECONOMIC AND ENVIRONMENTAL VIEW

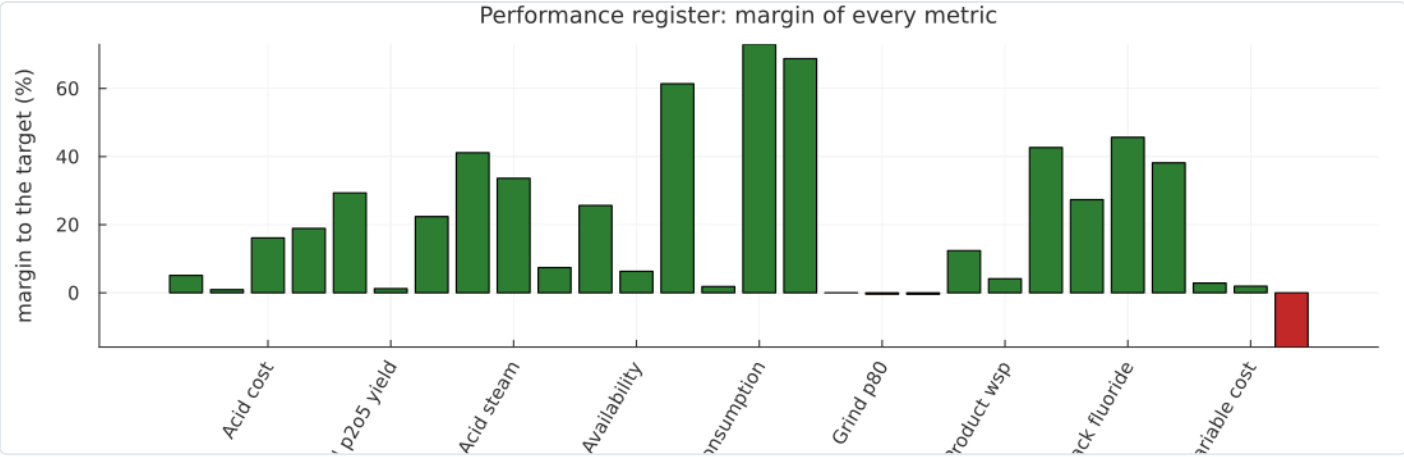
Quantity	Value
Product	1,712.9 kt of Dap at 46.05 % P2O5
Nitrogen	18.14 % (specification 17.4 to 19.2)
Moisture and solubility	1.75 % H2O and 88.5 % water-soluble P2O5
Ammonia	379.6 kt (0.3643 t per t P2O5, molar ratio 1.52)
Recycle ratio	7 : 1
Oversize and fines	6.6 % and 28.3 %
Specific energy	2.01 GJ per t P2O5 (56.8 kWh of power)
Dryer fuel	718.6 TJ of :natural_gas
Variable cost	671.81 M:USD (392.2 per t of product)
Revenue and margin	1.11614e6 M:USD of revenue, margin 651,204 per t
Carbon	56 kgCO2e per t P2O5 (scope 1 36,380 kt, scope 2 21,955 kt)
Stack	2.72 mg F per Nm3 (limit 5) and 21.8 mg dust per Nm3 (limit 30)
Water	3,522.8 thousand m3 (3.38 t per t P2O5)



Where the energy of the campaign goes and the specific energy of the product



The variable cost of the tonne of product, item by item



Margin of every performance metric of the register against its target

The performance register of the campaign

● COMPLIANT

25 metrics

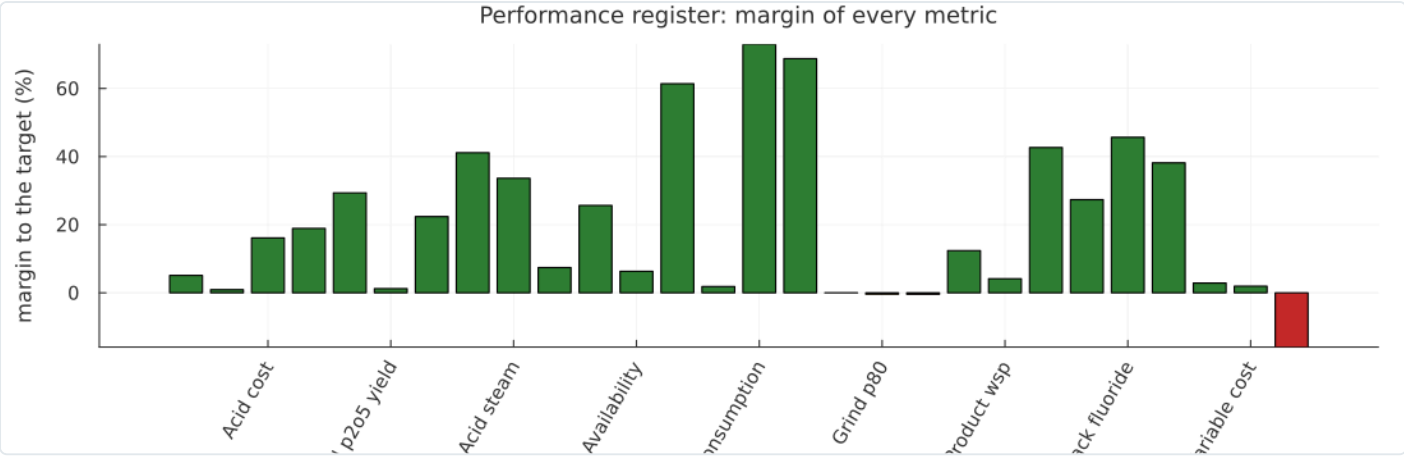
● AT RISK

2 metrics

● NONCOMPLIANT

1 metrics

Metric	Campaign	Target	Unit	Direction	Margin %	Status	Basis
:acid_concentration_yield	101.96	97	:pct	:ge	5.11	Compliant	P2O5 in the merchant acid over the P2O5 the evaporators concentrate: a closure check at three per cent of meter slack
:acid_consumption	2.82	2.85	:t_ph	:le	0.9321	Compliant	t H2SO4 per t P2O5, stoichiometric demand plus the registered excess
:acid_cost	503.33	600	:usd_per_t	:le	16.11	Compliant	variable cost per t P2O5 sold as merchant acid: feed by acid share, concentration in full
:acid_electricity	60.82	75	:kwh_per_t	:le	18.9	Compliant	kWh per t P2O5 concentrated, blower of the acid plant and evaporator set
:acid_f	0.3533	0.5	:wt_pct	:le	29.35	Compliant	fluorine of the merchant acid: the specification limit of the grade
:acid_p2o5_yield	97.68	96.5	:pct	:ge	1.22	Compliant	P2O5 reaching the filters over the P2O5 fed to the attack
:acid_so4	1.16	1.5	:wt_pct	:le	22.39	Compliant	sulphate of the merchant acid: the specification limit of the grade
:acid_solids	0.2943	0.5	:wt_pct	:le	41.13	Compliant	solids of the merchant acid: the specification limit of the grade
:acid_steam	0.4115	0.62	:t_ph	:le	33.63	Compliant	t steam per t P2O5 concentrated, at an economy of 3.0
:acid_strength	55.85	52	:wt_pct	:ge	7.41	Compliant	P2O5 in the merchant acid
:ammonia_consumption	0.3643	0.49	:t_ph	:le	25.66	Compliant	t NH3 per t P2O5 of DAP
:availability	97.81	92	:pct	:ge	6.31	Compliant	hours the reaction section was available
:carbon_intensity	55.98	145	:kg_ph	:le	61.39	Compliant	kgCO2e per t P2O5, scope 1 and 2
:completeness	99.81	98	:pct	:ge	1.85	Compliant	usable intervals of the historian
:electricity_consumption	56.79	210	:kwh_per_t	:le	72.96	Compliant	kWh per t P2O5, mine to bagged product
:filtration_rate	8.44	5	:t_per_m2_d	:ge	68.76	Compliant	t P2O5 per m2 of filter per day
:flotation_recovery	86.01	86	:pct	:ge	0.0075	Compliant	registered recoveries of the rougher-cleaner circuit
:grind_p80	150.6	150	:um	:le	-0.4012	At risk	liberation size of the blend
:plant_p2o5_recovery	83.65	84	:pct	:ge	-0.4154	At risk	P2O5 in the product plus the merchant acid over the P2O5 in the flotation feed
:product_moisture	1.75	2	:wt_pct	:le	12.39	Compliant	moisture of the bagged product
:product_wsp	88.51	85	:wt_pct	:ge	4.12	Compliant	water-soluble P2O5 of the product
:specific_energy	2.01	3.5	:gj_per_t	:le	42.66	Compliant	GJ per t P2O5: steam of the evaporators, fuel of the dryers and power
:stack_dust	21.8	30	:count	:le	27.35	Compliant	mg dust per Nm3 at the stack, the permitted limit
:stack_fluoride	2.72	5	:count	:le	45.65	Compliant	mg F per Nm3 at the stack, the permitted limit
:steam_consumption	0.402	0.65	:t_ph	:le	38.16	Compliant	t steam per t P2O5 fed, weak acid at 28.5 % raised to 52 % at an economy of 3.0
:sulphur_consumption	0.9227	0.95	:t_ph	:le	2.87	Compliant	t sulphur per t P2O5 at 99.7% conversion
:variable_cost	392.2	400	:usd_per_t	:le	1.95	Compliant	variable cost per tonne of DAP
:water_soluble_loss	2.32	2	:pct	:le	-15.93	Noncompliant	water-soluble P2O5 lost with the gypsum, per cent of the P2O5 fed



Margin of every performance metric of the register against its target

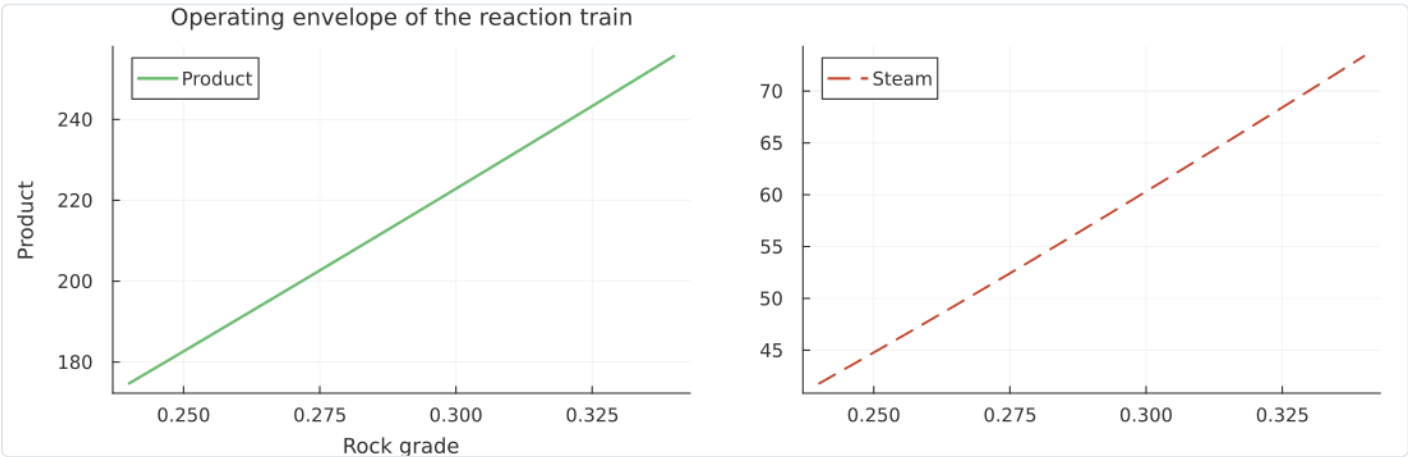
The digital twin: steady state, limits and alignment

The steady-state twin of the reaction train is a square system of 15 nonlinear balances solved with ModelingToolkit (ModelingToolkit nonlinear balance, mtkcompile + NonlinearProblem). It converged in 11.93 s with a largest residual of 0.0 -- the residual is reported because a solution nobody checked is a guess.

Unknown	Value	Unit	Description
:p2o5_fed	150	:t_ph	P2o5 fed
:excess	0.02562	:fraction	Excess
:free_so4	2.6606	:wt_pct	Free so4
:temperature	77.2418	:deg_c	Temperature
:efficiency	0.979187	:fraction	Efficiency
:p2o5_digested	146.878	:t_ph	P2o5 digested
:gypsum	737.2408	:t_ph	Gypsum
:p2o5_acid	142.4545	:t_ph	P2o5 acid
:water_liquor	366.0578	:t_ph	Water liquor
:acid_strength_weak	28.014	:wt_pct	Acid strength weak
:steam	65.1559	:t_ph	Steam
:water_evaporated	234.5613	:t_ph	Water evaporated
:ammonia	51.9605	:t_ph	Ammonia
:product	235.1041	:t_ph	Product
:merchant_p2o5	34.1891	:t_ph	Merchant p2o5

OPERATING WINDOW AT THE SOLUTION

Variable	At the solution	Low	High	Unit	Margin low	Margin high	Status
:acid_strength_weak	28.01	27	33	:wt_pct	1.01	4.99	Compliant
:ammonia	51.96	20	55	:t_ph	31.96	3.04	Compliant
:excess	0.0256	0.015	0.055	:fraction	0.0106	0.0294	Compliant
:free_so4	2.66	1.2	7	:wt_pct	1.46	4.34	Compliant
:steam	65.16	40	130	:t_ph	25.16	64.84	Compliant
:temperature	77.24	72	83	:deg_c	5.24	5.76	Compliant

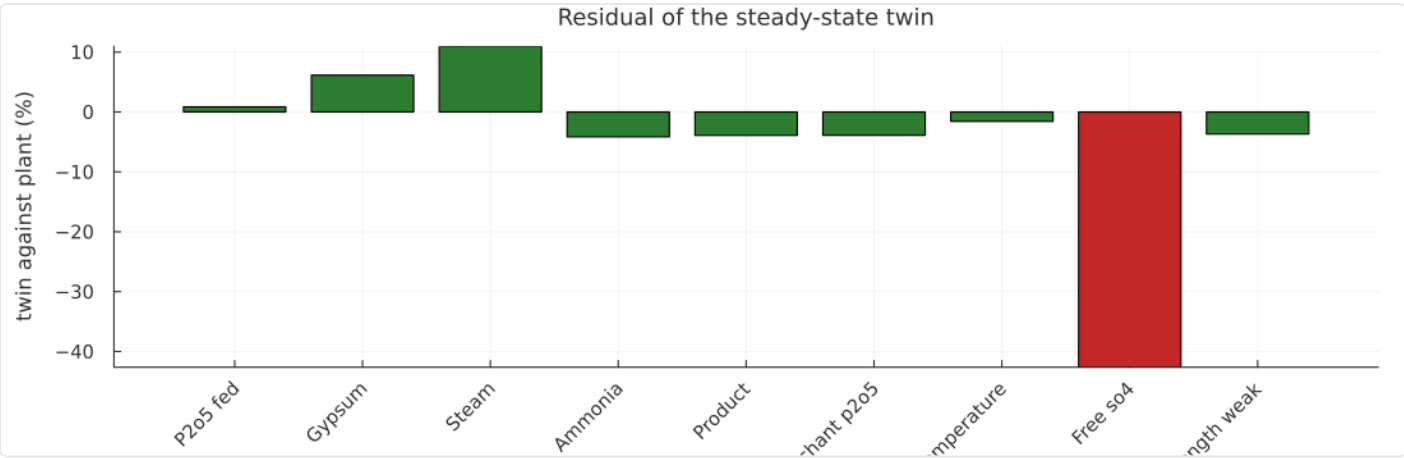


Operating envelope of the reaction train from the steady-state twin

Against the campaign it is supposed to describe, the twin scores 91.4 % of validity: the residuals below are what a commissioning would absorb as a bias, and the reason they are printed is that a twin is only as good as this table.

Variable	Twin	Plant	Unit	Residual %	Tolerance %	Status
:p2o5_fed	122.67	121.62	:t_ph	0.8598	10	Compliant
:gypsum	600.38	565.63	:t_ph	6.14	12	Compliant
:steam	54.25	48.89	:t_ph	10.96	25	Compliant
:ammonia	42.46	44.3	:t_ph	-4.16	25	Compliant
:product	192.11	199.92	:t_ph	-3.91	25	Compliant
:merchant_p2o5	27.94	29.07	:t_ph	-3.9	20	Compliant

Variable	Twin	Plant	Unit	Residual %	Tolerance %	Status
temperature	76.44	77.66	:deg_c	-1.57	5	Compliant
free_so4	1.56	2.72	:wt_pct	-42.62	15	Noncompliant
acid_strength_weak	27.77	28.84	:wt_pct	-3.68	8	Compliant



Residual of the steady-state twin against the campaign it describes

The dynamic twin and the state estimation

The dynamic twin integrates 11 states with OrdinaryDiffEqTsit5.Tsit5(typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!), FastBroadcast.Serial)(OrdinaryDiffEqCore.trivial_limiter!, OrdinaryDiffEqCore.trivial_limiter!, FastBroadcast.Serial()) on the hourly clock of the plant, from the same correlations the historian uses: the holdup of the attack tanks, the strength of the acid, the moisture of the granulator bed, the holdup of the mills and the fouling of the evaporator tubes.

DYNAMIC TRAJECTORY OF THE TWIN

Quantity	Value
Horizon	0.0 to 48.0 h
Status	compliant (Success)
P2O5 fed over the horizon	3,934.9 t
Liquor of the attack tanks	306 t to 306 t (-0.0009 %)
Highest temperature	78.02 C
Weakest acid	51.6 % P2O5
Fouling at the end	0.629
Integration time	4.55 s

The Kalman filter of the twin walked the historian for 336 intervals and estimated the states nobody measures. The innovation of every measurement is the table below: it is the residual the controller sees, and a sensor that drifts shows up here first.

Measurement	RMSE of the innovation	Bias	Intervals
:acid_strength	2.21	-2.15	336
:acid_strength_weak	40.48	40.1	336
:bed_moisture	1.4	1.4	336
:filter_rate	58.93	-58.8	336
:fouling_pct	0	0	336
:free_so4	27.28	-26.88	336
:granulator_temperature	1.05	-0.6328	336
:mill_inventory	142.33	-140.42	336
:mill_power	1,874.03	-1,847.63	336
:p80	51.84	-50.83	336
:product_rate	1,283.84	-1,281.18	336
:temperature	15.96	-0.7315	336

Predictive control of the reaction train

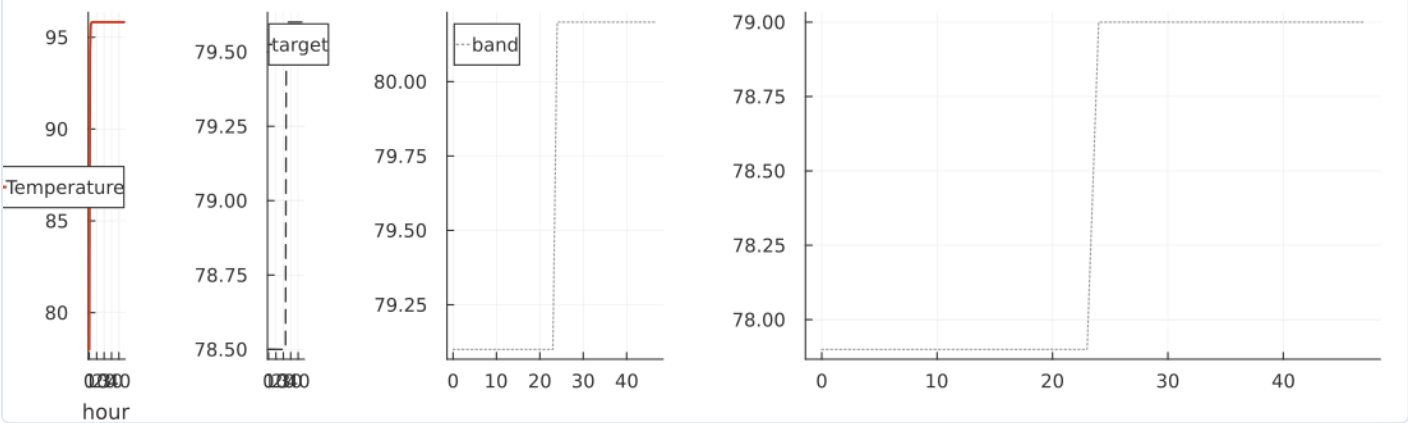
The controller is the one the plant would run: a linear predictive controller over the discrete model 11 x 11 obtained by linearising the ODE twin at its operating point (12 prediction steps, 4 control steps, sample time 1 h), with a Kalman filter estimating the states from the measurements. The plant of the test is the *nonlinear* twin, so the controller never sees the equations it is controlling.

CLOSED LOOP OF THE TWO CONTROLLERS

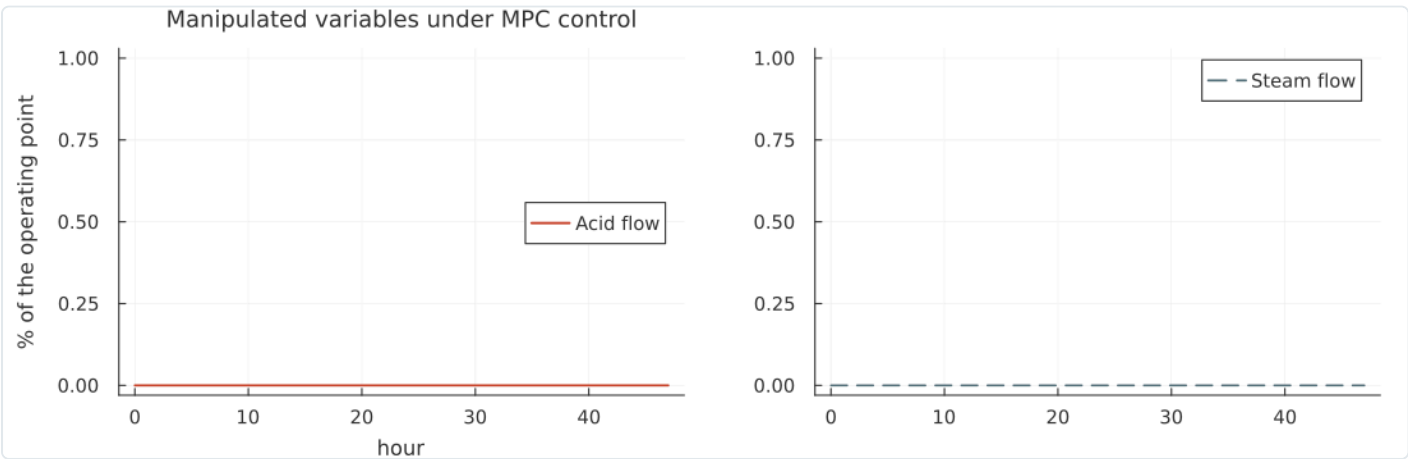
Quantity	Value
Scenario	:setpoint over 48 hours
Model	11 states, 6 inputs, 12 outputs, Ts = 1 h
Controllability	conditioning 3.4124288612524188e16
Poles	largest magnitude 1
MPC error	14,863 bands and 284 violations
PI error	2,867.68 bands and 233 violations

Controlled variable	Band	MPC error	PI error	Improvement %	MPC worst	PI worst	MPC hours out	PI hours out
:temperature	0.6	1,310.92	41.86	-3,031.32	28.87	0.9654	47	0
:free_so4	0.2	8,282.93	604.12	-1,271.06	195.69	17.27	47	42
:acid_strength_weak	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:acid_strength	0.15	4,041	942.67	-328.68	89.91	23.24	48	48
:product_rate	6	1,102.69	1,115.28	1.13	23.35	23.35	48	48
:filter_rate	0.35	70.65	60.84	-16.12	1.48	1.27	48	48
:mill_power	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:p80	6	54.82	102.9	46.73	1.17	2.19	46	47
:bed_moisture	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:granulator_temperature	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:mill_inventory	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
:fouling_pct	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

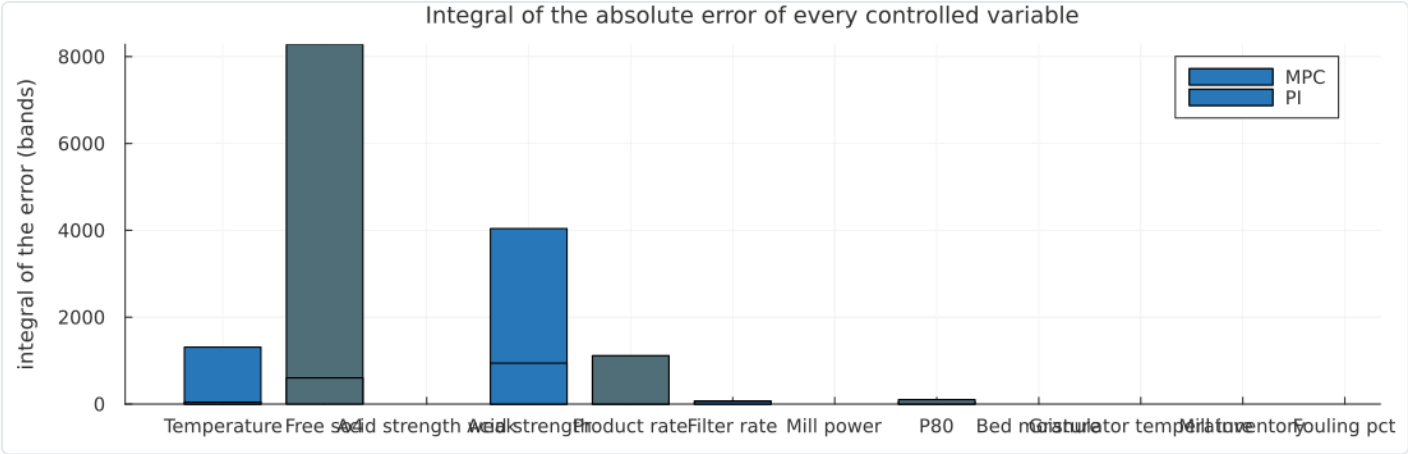
Temperature under MPC control (:setpoint)



Closed loop of the predictive controller



Manipulated variables the predictive controller used



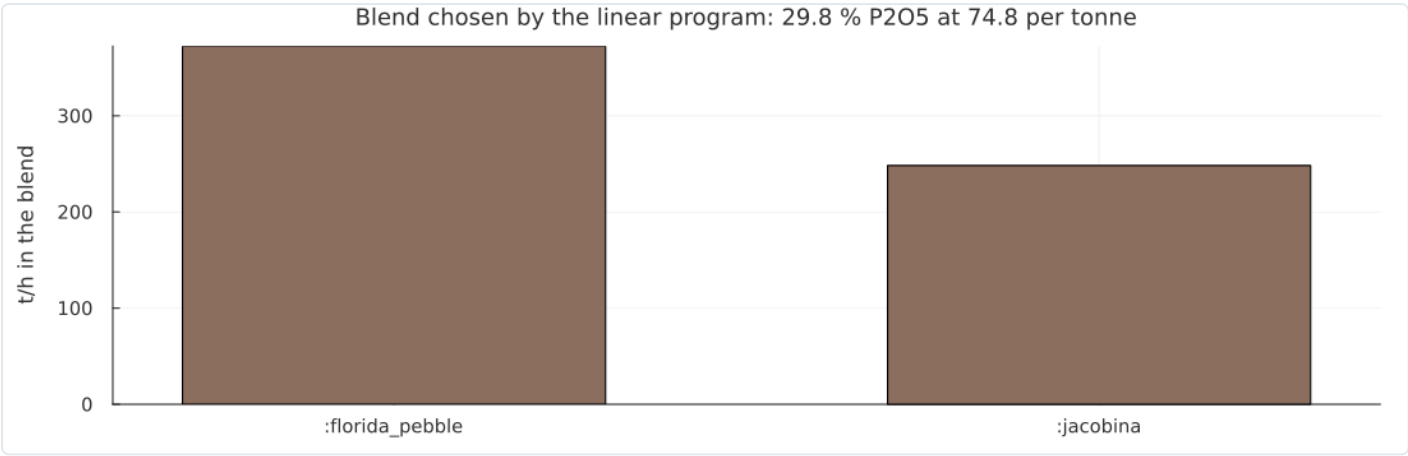
Integral of the absolute error of the two controllers, output by output

WORK THE TWO CONTROLLERS ASKED OF THE PLANT

Manipulated variable	MPC movement %	PI movement %	MPC reversals	PI reversals	MPC range %	PI range %	
:acid_flow		0	14.94	0	0	0	14.94
:ammonia_flow		0	0	0	0	0	0
:cooling_water		0	40	0	0	0	40
:mill_feed		0	2.62	0	0	0	2.62
:steam_flow		0	17.75	0	0	0	17.75
:wash_water		0	0	0	0	0	0

Optimisation of the blend, the operating point and the sourcing

Three models of the same plant are solved every day: the blend of the ore the digester is fed -- a linear program over the registered assays with the specifications of the reaction section --, the operating point of the reaction train -- a nonlinear program over the correlations of the twin -- and the annual sourcing plan, a mixed-integer program because a body is bought by the shipload.



Ore blend the optimiser chose

OPTIMAL BLEND (HIGHS, STATUS :COMPLIANT, 29.8 % P2O5 AT 74.8 PER T)

Ore body	t/h	Share %	Price	P2O5 %	CaO:P2O5	Fe+Al %	MgO %
:florida_pebble	372.41	60	78	29.8	1.53	2.5	0.5
:jacobina	248.28	40	70	29.8	1.54	7	1.5
:boucraa	0	0	96	34.8	1.47	0.6	0.4
:gafsa	0	0	74	29	1.62	1.7	1.1
:khouribga	0	0	82	30.9	1.63	0.8	0.6
:tapira	0	0	88	33.4	1.44	5	1

WHAT ONE MORE UNIT OF EACH SPECIFICATION IS WORTH

Specification	Shadow price
:mgo	-800
:fe_al	-0
:p2o5	0
:sio2	-0
:cao_p2o5	-0
:co2	-0

OPERATING POINT (IPOPT, STATUS :COMPLIANT, MARGIN 184.7 PER T)

Nature	Variable	Optimum
:decision	:recycle_acid_flow	450
:decision	:cooling_water_flow	248
:decision	:steam_flow	80.82
:decision	:rock_feed	523.81
:decision	:wash_water_flow	220.19
:decision	:ammonia_flow	57.05
:decision	:sulphuric_acid_flow	485.42
:controlled	:product_moisture	1.45
:controlled	:acid_strength_weak	27
:controlled	:gypsum_t	674.99
:controlled	:sulfate_ratio	2.3
:controlled	:p2o5_product	124.18
:controlled	:steam_required	80.82
:controlled	:product_wsp	88.29
:controlled	:gypsum_free_p2o5	0.5268
:controlled	:acid_excess	1.5
:controlled	:reactor_temperature	80.28
:controlled	:p2o5_fed	167.99

Nature	Variable	Optimum
:controlled	:filtration_rate	12.52
:controlled	:product_n	17.4
:controlled	:acid_strength	53.37

ANNUAL SOURCING PLAN (HIGHS, 72 BINARY DECISIONS, 253.45 PER T P2O5)

Ore body	t per year	Share %	Price	Cost	Months	P2O5 %
:gafsa	1.283079e6	27.8	74	9.4947875e7	4	29
:florida_pebble	1.248634e6	27.05	78	9.7393476e7	4	29.8
:jacobina	1.248634e6	27.05	70	8.7404401e7	4	29.8
:tapira	835,538	18.1	88	7.3527364e7	3	33.4
:boucraa	0	0	96	0	0	34.8
:khouribga	0	0	82	0	0	30.9

The inferential sensors: process, vision and acoustics

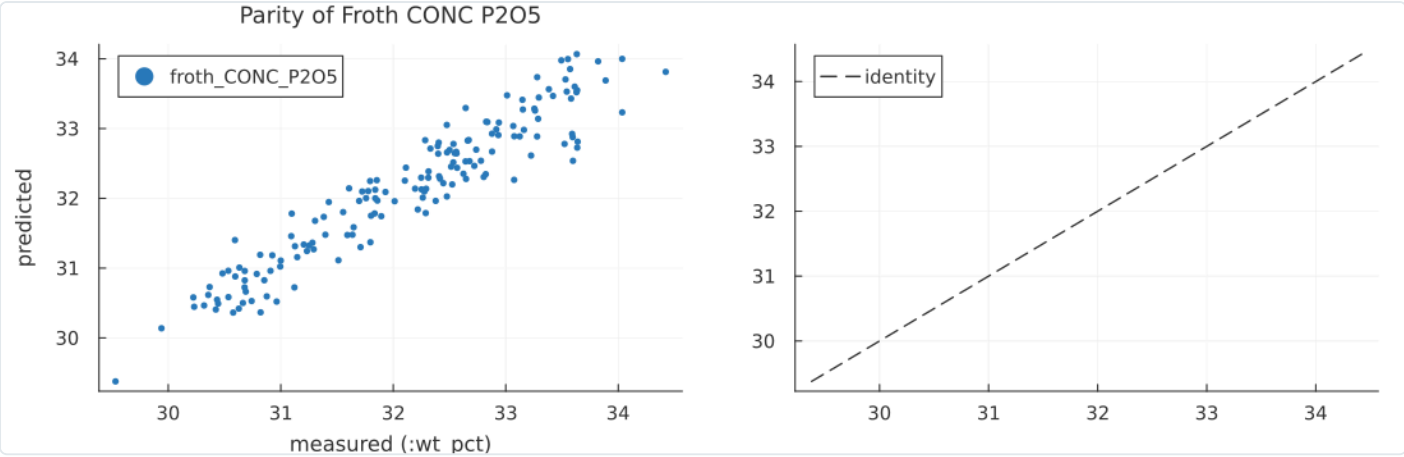
THE REGISTER OF THE INFERENTIAL SENSORS

Quantity	Value									
Sensors	14 inferential models, of which 4 read an image or a sound									
Median R2	0.57364									
Best sensor	:pump_PUMP_VIBRATION									
Status	:at_risk									

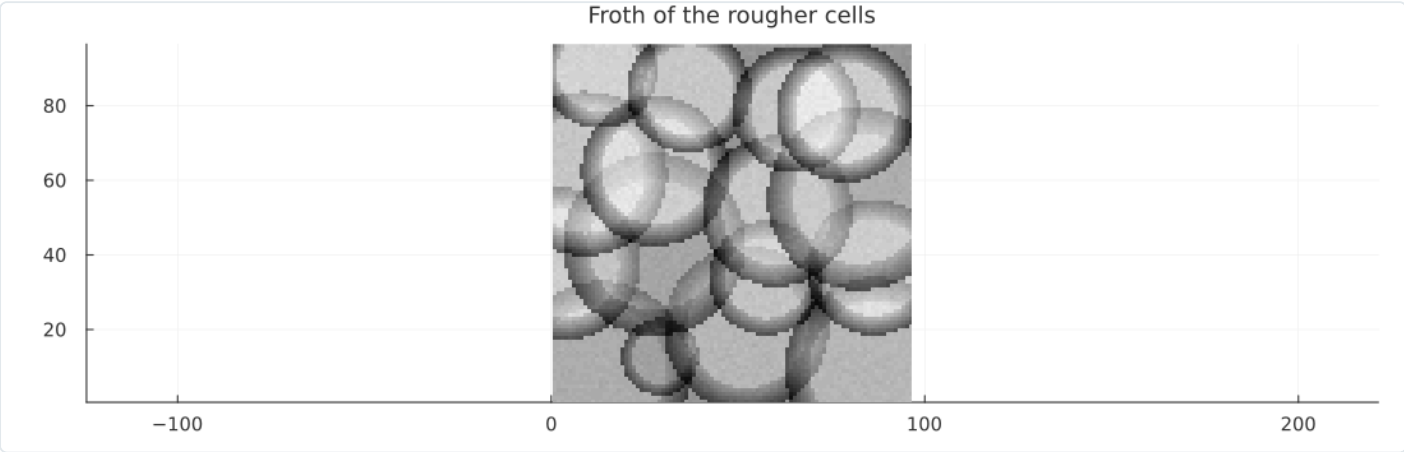
Sensor	Infers	Modality	Unit	R2	RMSE	Bias	In domain %	Drift	Status	Inputs
:pump_PUMP_VIBRATION	:PUMP_VIBRATION	:acoustic	:count	0.9896	0.073906	-0.0097	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:mill_MILL_POWER	:MILL_POWER	:acoustic	:kw	0.9423	83.7837	-23.52	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:process_MILL_POWER	:MILL_POWER	:process	:kw	0.93678	106.9652	3.45	100	1.38	Healthy	ROM_FEED, SLURRY_DENSITY, MILL_FEED_WATER
:process_STEAM_FLOW	:STEAM_FLOW	:process	:t_ph	0.93115	1.6977	-0.0218	100	-0.0087	Healthy	EVAP_FEED_FLOW, STRONG_ACID_P2O5, EVAP_VACUUM
:process_CONC_P2O5	:CONC_P2O5	:process	:wt_pct	0.91007	0.30973	0.0084	100	0.0034	Healthy	FLOT_FEED_P2O5, CYCLONE_P80, REAGENT_FLOW
:froth_CONC_P2O5	:CONC_P2O5	:visual	:wt_pct	0.89295	0.356853	-0.0189	n/a	0	Healthy	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:process	:wt_pct	0.69479	0.175129	0.0023	100	0.0009	Healthy	SLURRY_SG, ATTACK_TEMP, FREE_SO4, WASH_WATER_FLOW
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:process	:wt_pct	0.45249	0.19828	0.0007	100	0.0003	Healthy	EVAP_TEMP, STEAM_FLOW, EVAP_VACUUM
:rock_CYCLONE_P80	:CYCLONE_P80	:visual	:um	0.39939	3.044	0.5433	n/a	0	Degraded	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_PRODUCT_N	:PRODUCT_N	:process	:wt_pct	0.33168	0.2647	0.0021	100	0.0009	Healthy	AMMONIA_FLOW, PRODUCT_FLOW, MELT_DENSITY
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:process	:wt_pct	0.21885	0.018076	0.0001	100	0	Healthy	FILTER_RATE, FILTER_VACUUM, WASH_WATER_FLOW, WEAK_ACID_P2O5
:process_FILTER_RATE	:FILTER_RATE	:process	:t_per_m2_d	0.21477	1.165	0.0177	100	0.0071	Healthy	FILTER_VACUUM, WEAK_ACID_P2O5, SLURRY_SG, GYPSUM_FREE_P2O5
:process_CYCLONE_P80	:CYCLONE_P80	:process	:um	0.03116	6.1781	-0.1283	100	-0.0513	Healthy	MILL_POWER, SLURRY_DENSITY, CYCLONE_PRESSURE, MILL_FEED_WATER
:process_ATTACK_TEMP	:ATTACK_TEMP	:process	:deg_c	-4.0e-5	16.3148	0.2679	100	0.1072	Healthy	H2SO4_FLOW, ROCK_FEED, RECYCLE_ACID_FLOW, SLURRY_SG

WHAT THE SENSORS SAY AGAINST WHAT THE PLANT MEASURED

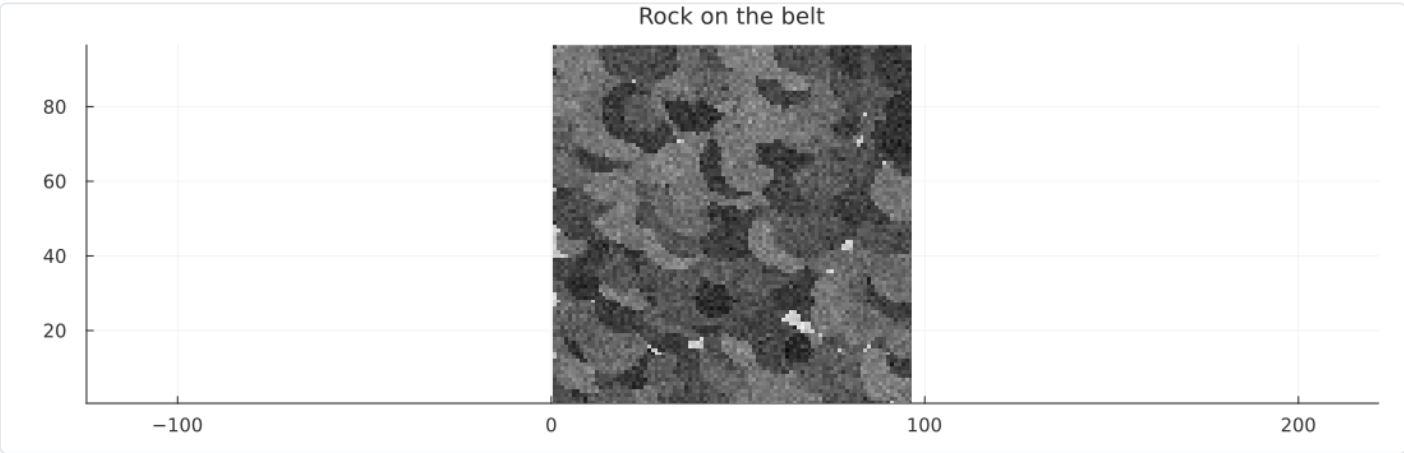
Sensor	Infers	Unit	Estimate	Measured	Difference %	RMSE	Status
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:wt_pct	55.77	55.85	-0.1382	0.1983	Compliant
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:wt_pct	28.75	28.7	0.1628	0.1751	Compliant
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:wt_pct	0.5028	0.5069	-0.8103	0.0181	Compliant
:process_CYCLONE_P80	:CYCLONE_P80	:um	150.82	149.52	0.8702	6.18	Compliant
:process_MILL_POWER	:MILL_POWER	:kw	5,523.06	5,524.95	-0.0341	106.97	Compliant
:process_PRODUCT_N	:PRODUCT_N	:wt_pct	18.11	18.11	-0.0093	0.2647	Compliant
:process_FILTER_RATE	:FILTER_RATE	:t_per_m2_d	8.76	8.89	-1.46	1.16	Compliant
:process_CONC_P2O5	:CONC_P2O5	:wt_pct	32.4	32.47	-0.2148	0.3097	Compliant
:process_ATTACK_TEMP	:ATTACK_TEMP	:deg_c	77.55	78.18	-0.8028	16.31	Compliant
:process_STEAM_FLOW	:STEAM_FLOW	:t_ph	51.47	51.69	-0.4197	1.7	Compliant



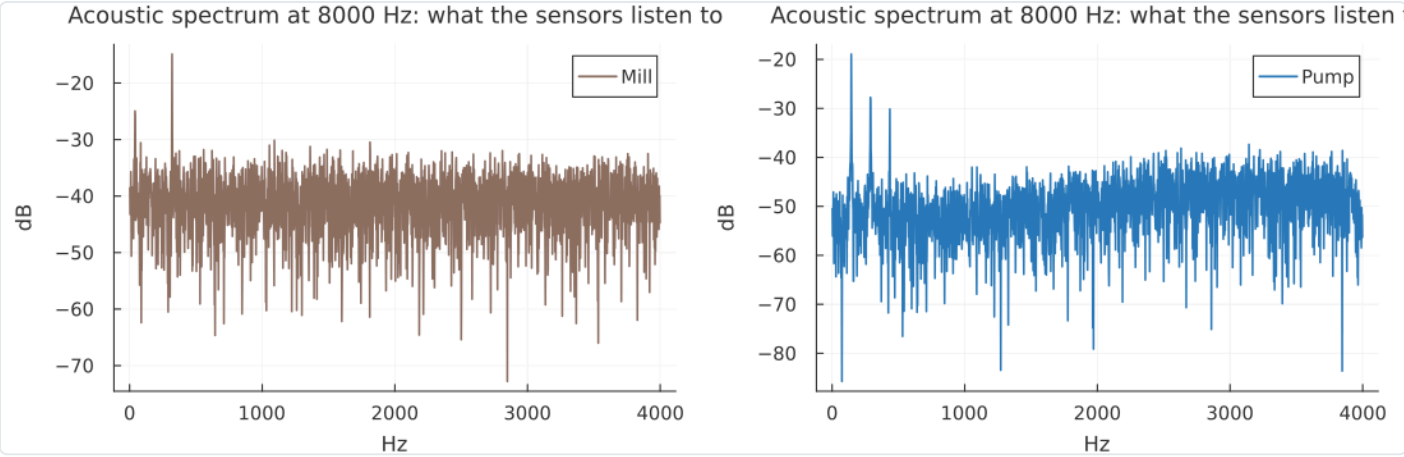
Parity of the froth sensor against the analyser



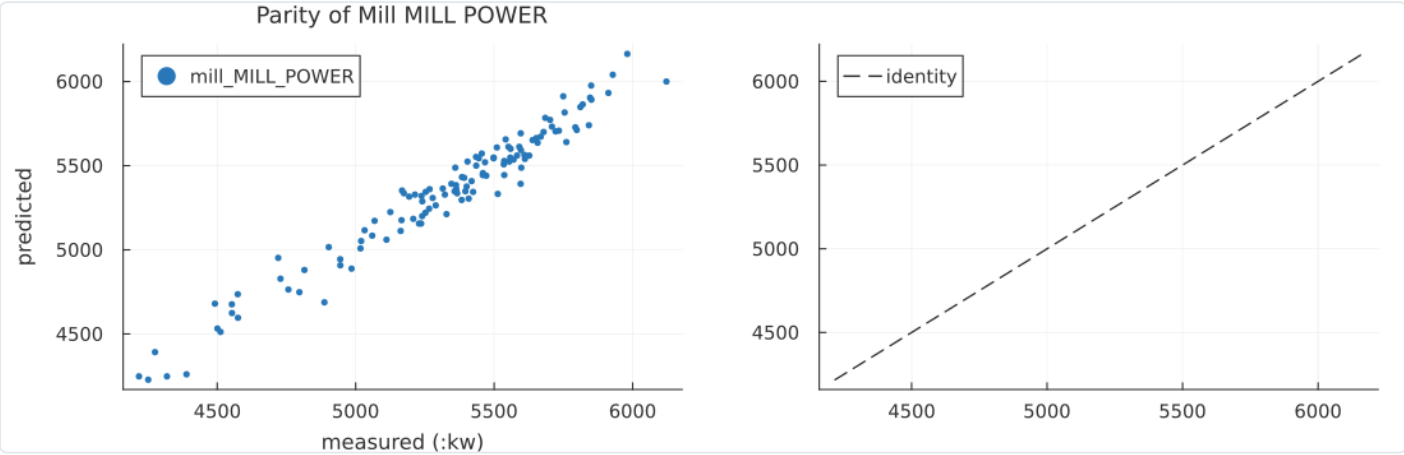
The froth image the visual sensor reads the grade from



The image of the rock the visual sensor reads the grind from



Spectra of the mill and of the pump: the two acoustic channels of the plant



Parity of the acoustic sensor of the mill against the power it draws

Fault detection and diagnostics

DIAGNOSTIC SUMMARY

Quantity	Value
Findings	100 (30 critical, 65 warning, 5 info)
Value at stake	1.67734421e8 :USD a year
Hours in alarm	19,447 h
Rules applied	20 rules of the catalogue
Deviations caught	8 of 15 (53.3 %)
Status	:action_required

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	27.63	29.5	:wt_pct	6.34	2026-12-06T10:00:00	2026-12-11T03:00:00	114	1.76516e7
Critical	:flotation	:concentrate_grade_low	:CONC_P2O5	28.6	29.5	:wt_pct	3.07	2026-03-13T10:00:00	2026-03-21T00:00:00	183	8.527489e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.64	50.5	:wt_pct	3.68	2026-12-06T12:00:00	2026-12-11T03:00:00	112	3.815317e6
Critical	:evaporation	:acid_strength_low	:STRONG_ACID_P2O5	48.74	50.5	:wt_pct	3.49	2026-03-13T12:00:00	2026-03-16T03:00:00	64	3.619393e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7723	1.2	:wt_pct	35.64	2026-12-06T11:00:00	2026-12-11T00:00:00	110	3.226419e6
Critical	:attack	:sulfate_deficit	:FREE_SO4	0.7919	1.2	:wt_pct	34.01	2026-03-13T11:00:00	2026-03-16T01:00:00	63	3.079024e6
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.3	5	:count	5.98	2026-06-28T07:00:00	2026-06-28T16:00:00	10	574,052
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.24	5	:count	4.76	2026-06-18T01:00:00	2026-06-18T15:00:00	15	456,933
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.68	2026-06-22T15:00:00	2026-06-23T01:00:00	11	449,616
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.6	2026-06-30T21:00:00	2026-07-01T23:00:00	27	441,632
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.56	2026-06-21T14:00:00	2026-06-22T09:00:00	20	437,807
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.23	5	:count	4.53	2026-06-24T13:00:00	2026-06-24T19:00:00	7	435,166
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.21	5	:count	4.22	2026-06-28T22:00:00	2026-06-29T13:00:00	16	405,194
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.2	5	:count	3.94	2026-06-19T10:00:00	2026-06-19T17:00:00	8	377,932
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.19	5	:count	3.8	2026-06-16T05:00:00	2026-06-17T07:00:00	27	364,612
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.18	5	:count	3.52	2026-06-23T12:00:00	2026-06-23T19:00:00	8	338,395
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.18	5	:count	3.51	2026-06-17T13:00:00	2026-06-17T18:00:00	6	337,031
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.17	5	:count	3.44	2026-06-26T18:00:00	2026-06-27T00:00:00	7	330,181
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.16	5	:count	3.11	2026-06-19T21:00:00	2026-06-21T10:00:00	38	298,491
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.13	5	:count	2.54	2026-06-24T21:00:00	2026-06-26T05:00:00	33	243,618
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.13	5	:count	2.51	2026-06-30T09:00:00	2026-06-30T12:00:00	4	240,763
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.31	2026-06-29T16:00:00	2026-06-30T05:00:00	14	222,178
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.12	5	:count	2.3	2026-06-27T14:00:00	2026-06-27T16:00:00	3	221,260
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.09	5	:count	1.86	2026-06-26T11:00:00	2026-06-26T16:00:00	6	178,928
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.34	17.4	:wt_pct	6.09	2026-12-06T08:00:00	2026-12-11T06:00:00	119	145,116
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.07	5	:count	1.49	2026-06-23T21:00:00	2026-06-24T02:00:00	6	142,967
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.39	17.4	:wt_pct	5.83	2026-03-13T08:00:00	2026-03-16T05:00:00	70	139,058
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.05	5	:count	1.07	2026-06-19T01:00:00	2026-06-19T03:00:00	3	103,086
Critical	:granulation	:product_nitrogen_low	:PRODUCT_N	16.96	17.4	:wt_pct	2.51	2026-08-09T07:00:00	2026-08-20T03:00:00	261	59,921
Critical	:filtration	:stack_fluoride_high	:STACK_F	5.02	5	:count	0.3091	2026-06-24T05:00:00	2026-06-24T07:00:00	3	29,673
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.32	4	t_per_m2_d	16.92	2026-12-06T16:00:00	2026-12-11T00:00:00	105	4.1379752e7
Warning	:filtration	:filter_rate_low	:FILTER_RATE	3.35	4	t_per_m2_d	16.27	2026-03-13T14:00:00	2026-03-16T00:00:00	59	3.9785148e7
Warning	:beneficiation	:grind_coarse	:CYCLONE_P80	190.9	185	:um	3.19	2026-03-16T22:00:00	2026-03-21T02:00:00	101	8.344416e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.77	5.6	:count	14.74	2026-12-06T06:00:00	2026-12-11T02:00:00	117	1.189036e6
Warning	:granulation	:scrubber_ph_low	:SCRUBBER_PH	4.8	5.6	:count	14.2	2026-03-13T04:00:00	2026-03-16T02:00:00	71	1.144873e6

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,428.17	4,600	:kw	18	2026-11-30T02:00:00	2026-12-06T11:00:00	154	563,154
Warning	:beneficiation	:slurry_density_high	:SLURRY_DENSITY	44.4	44	:wt_pct	0.9005	2026-10-12T01:00:00	2026-10-16T00:00:00	96	560,460
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,421.56	4,600	:kw	17.86	2026-12-28T01:00:00	2026-12-31T23:00:00	95	558,661
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,415.4	4,600	:kw	17.73	2026-04-06T01:00:00	2026-04-12T20:00:00	164	554,475
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,414.21	4,600	:kw	17.7	2026-01-01T05:00:00	2026-01-04T21:00:00	89	553,661
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,412.27	4,600	:kw	17.66	2026-10-12T02:00:00	2026-10-18T21:00:00	164	552,347
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,411.47	4,600	:kw	17.64	2026-12-14T02:00:00	2026-12-20T21:00:00	164	551,798
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,410.44	4,600	:kw	17.62	2026-11-23T02:00:00	2026-11-29T20:00:00	163	551,101
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,409.68	4,600	:kw	17.6	2026-02-02T01:00:00	2026-02-08T21:00:00	165	550,584
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,407.27	4,600	:kw	17.55	2026-01-12T01:00:00	2026-01-18T21:00:00	165	548,943
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,405.7	4,600	:kw	17.52	2026-02-23T02:00:00	2026-03-01T21:00:00	164	547,873
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,404.54	4,600	:kw	17.49	2026-04-13T02:00:00	2026-04-19T20:00:00	163	547,089
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,403.94	4,600	:kw	17.48	2026-01-05T02:00:00	2026-01-11T21:00:00	164	546,677
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,402.83	4,600	:kw	17.45	2026-11-02T02:00:00	2026-11-08T20:00:00	163	545,925
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,402.37	4,600	:kw	17.44	2026-11-16T01:00:00	2026-11-22T20:00:00	164	545,609
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.91	4,600	:kw	17.43	2026-04-27T01:00:00	2026-05-03T20:00:00	164	545,300
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.8	4,600	:kw	17.43	2026-09-07T02:00:00	2026-09-13T21:00:00	164	545,225
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.78	4,600	:kw	17.43	2026-03-30T01:00:00	2026-04-05T21:00:00	165	545,210
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,401.17	4,600	:kw	17.42	2026-11-09T02:00:00	2026-11-15T21:00:00	164	544,793
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.73	4,600	:kw	17.41	2026-05-18T01:00:00	2026-05-24T20:00:00	164	544,496
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.47	4,600	:kw	17.4	2026-09-14T01:00:00	2026-09-20T21:00:00	165	544,322
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.26	4,600	:kw	17.4	2026-01-26T02:00:00	2026-02-01T21:00:00	164	544,176
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,400.14	4,600	:kw	17.39	2026-05-04T02:00:00	2026-05-10T22:00:00	165	544,096
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,399.97	4,600	:kw	17.39	2026-02-16T02:00:00	2026-02-22T20:00:00	163	543,981
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.98	4,600	:kw	17.37	2026-03-23T01:00:00	2026-03-29T21:00:00	165	543,306
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.8	4,600	:kw	17.37	2026-10-05T01:00:00	2026-10-11T21:00:00	165	543,187
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,398.46	4,600	:kw	17.36	2026-05-25T01:00:00	2026-05-31T21:00:00	165	542,955
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.73	4,600	:kw	17.34	2026-04-20T02:00:00	2026-04-26T21:00:00	164	542,458
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.56	4,600	:kw	17.34	2026-02-09T01:00:00	2026-02-15T20:00:00	164	542,344
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.44	4,600	:kw	17.34	2026-12-21T01:00:00	2026-12-27T21:00:00	165	542,257
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,397.44	4,600	:kw	17.34	2026-09-28T01:00:00	2026-10-04T21:00:00	165	542,257
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,395.55	4,600	:kw	17.29	2026-09-21T01:00:00	2026-09-27T21:00:00	165	540,977
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,393.98	4,600	:kw	17.26	2026-03-09T01:00:00	2026-03-13T11:00:00	107	539,910
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,390.52	4,600	:kw	17.19	2026-10-19T01:00:00	2026-10-25T21:00:00	165	537,552
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.85	4,600	:kw	17.17	2026-03-02T01:00:00	2026-03-08T21:00:00	165	537,097
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.3	4,600	:kw	17.16	2026-10-26T01:00:00	2026-11-01T21:00:00	165	536,724
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,389.12	4,600	:kw	17.15	2026-01-19T01:00:00	2026-01-25T21:00:00	165	536,601
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,385.71	4,600	:kw	17.08	2026-05-11T00:00:00	2026-05-17T20:00:00	165	534,280
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,381.92	4,600	:kw	17	2026-08-31T06:00:00	2026-09-06T20:00:00	159	531,706
Warning	:beneficiation	:slurry_density_high	:SLURRY_DENSITY	44.36	44	:wt_pct	0.8293	2026-10-06T11:00:00	2026-10-11T23:00:00	133	516,191
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,336.31	4,600	:kw	16.01	2026-07-06T03:00:00	2026-07-19T16:00:00	326	500,689
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,277.65	4,600	:kw	14.73	2026-12-11T13:00:00	2026-12-13T21:00:00	57	460,803
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,244.57	4,600	:kw	14.01	2026-06-29T06:00:00	2026-07-05T19:00:00	158	438,308
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,244.52	4,600	:kw	14.01	2026-06-08T06:00:00	2026-06-14T16:00:00	155	438,276
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.56	4,600	:kw	13.64	2026-08-10T06:00:00	2026-08-16T16:00:00	155	426,738

Severity	Area	Rule	Signal	Measured	Reference	Unit	Deviation %	From	To	Hours	Value at stake
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.04	4,600	:kw	13.63	2026-06-15T06:00:00	2026-06-21T16:00:00	155	426,389
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,227.04	4,600	:kw	13.63	2026-08-17T06:00:00	2026-08-23T16:00:00	155	426,388
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,225.6	4,600	:kw	13.6	2026-08-03T06:00:00	2026-08-09T15:00:00	154	425,410
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,223.12	4,600	:kw	13.55	2026-07-20T06:00:00	2026-07-26T15:00:00	154	423,719
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,220.2	4,600	:kw	13.48	2026-08-24T06:00:00	2026-08-30T16:00:00	155	421,739
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,219.38	4,600	:kw	13.46	2026-07-27T06:00:00	2026-08-02T16:00:00	155	421,179
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,217.93	4,600	:kw	13.43	2026-06-22T06:00:00	2026-06-28T15:00:00	154	420,192
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	5,209.92	4,600	:kw	13.26	2026-06-01T01:00:00	2026-06-07T16:00:00	160	414,748
Warning	:beneficiation	:mill_power_excess	:MILL_POWER	4,898.23	4,600	:kw	6.48	2026-03-16T19:00:00	2026-03-22T21:00:00	147	202,798
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.28	19.2	:wt_pct	0.4204	2026-10-30T18:00:00	2026-11-01T08:00:00	39	161,276
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.28	19.2	:wt_pct	0.4041	2026-10-26T14:00:00	2026-10-29T03:00:00	62	155,009
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.26	19.2	:wt_pct	0.3373	2026-10-25T10:00:00	2026-10-26T01:00:00	16	129,379
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.26	19.2	:wt_pct	0.3141	2026-10-29T11:00:00	2026-10-30T15:00:00	29	120,479
Warning	:granulation	:product_nitrogen_high	:PRODUCT_N	19.25	19.2	:wt_pct	0.2402	2026-11-01T16:00:00	2026-11-01T23:00:00	8	92,140
Warning	:utilities	:pump_vibration_high	:PUMP_VIBRATION	6.82	6	:count	13.59	2026-07-21T04:00:00	2026-07-31T00:00:00	237	1,019.23
Info	:filtration	:vacuum_low	:FILTER_VACUUM	163.88	200	:mmwc	18.06	2026-12-06T06:00:00	2026-12-11T01:00:00	116	173,375
Info	:filtration	:vacuum_low	:FILTER_VACUUM	165.18	200	:mmwc	17.41	2026-03-13T06:00:00	2026-03-16T01:00:00	68	167,126
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	547.66	336	:l_ph	63	2026-01-01T05:00:00	2026-03-13T08:00:00	1,708	122,766
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	545.85	336	:l_ph	62.46	2026-12-11T13:00:00	2026-12-31T23:00:00	491	121,714
Info	:flotation	:reagent_demand_high	:REAGENT_FLOW	537.6	336	:l_ph	60	2026-03-16T13:00:00	2026-12-06T07:00:00	6,355	116,926

DID THE DETECTOR CATCH WHAT THE CAMPAIGN INJECTED?

Injected deviation	Role	Kind	Magnitude	Unit	Hours	Detected	Rule that caught it	Findings
:cooling_fouling	:reactor_temperature	:drift	6.2	:deg_c	264	0	:none	0
:sulfate_deficit	:sulfate_ratio	:bias	-1.7	:wt_pct	288	0	:none	0
:grind_coarse	:grind_p80	:bias	42	:um	192	1	:grind_coarse	1
:reagent_starvation	:concentrate_grade	:bias	-1.5	:wt_pct	264	1	:reagent_demand_high	1
:filter_cloth_blinding	:filtration_rate	:drift	-1.9	:t_per_m2_d	168	0	:none	0
:evaporator_fouling	:steam_flow	:drift	13.5	:t_ph	360	0	:none	0
:acid_strength_loss	:acid_strength	:drift	-1.6	:wt_pct	312	0	:none	0
:scrubber_breakthrough	:stack_fluoride	:bias	2.4	:count	384	1	:stack_fluoride_high	21
:mill_liner_wear	:mill_load	:drift	430	:kw	264	1	:mill_power_excess	2
:pump_cavitation	:pump_vibration	:bias	3.6	:count	240	1	:pump_vibration_high	1
:sparger_blockage	:product_n	:bias	-1.2	:wt_pct	264	1	:product_nitrogen_low	1
:crusher_gap_drift	:oversize_pct	:bias	6.5	:pct	264	0	:none	0
:rock_grade_drift	:rock_grade	:drift	1.3	:wt_pct	192	0	:none	0
:cyclone_roping	:slurry_density	:bias	6.4	:wt_pct	240	1	:slurry_density_high	2
:ammonia_overfeed	:product_n	:bias	1.1	:wt_pct	192	1	:product_nitrogen_high	5

The catalogue of the detector holds 20 rules, each with the signal it watches, the reference it compares against and the window it averages over; 'PHOSPHATE.RULE_CATALOGUE' prints them and the notebook shows them rule by rule.

Compliance register and corrective actions

● COMPLIANCE

88.1 % of clauses

42 clauses in 9 frameworks

● COMPLIANT

37 clauses

● AT RISK

0 clauses

within 10 % of the limit

● NON-COMPLIANT

5 clauses

outside the limit

THE SCORECARD, FRAMEWORK BY FRAMEWORK

Framework	Title	Clauses	Compliant	At risk	Non-compliant	Compliant %
:eu_ied	EU Industrial Emissions Directive 2010/75	4	4	0	0	100
:ghg_protocol	GHG Protocol corporate standard	4	2	0	2	50
:ifa_bat	IFA/EFMA BAT for phosphate fertilisers	6	4	0	2	66.67
:iso_14001	ISO 14001:2015 environmental management	5	5	0	0	100
:iso_46001	ISO 46001:2019 water efficiency	3	3	0	0	100
:iso_50001	ISO 50001:2018 energy management	6	5	0	1	83.33
:iso_9001	ISO 9001:2015 quality management	8	8	0	0	100
:phosphogypsum_code	Phosphogypsum stack management code	3	3	0	0	100
:us_epa_40cfr63	US EPA 40 CFR 63 subpart AA	3	3	0	0	100

Framework	Clause	Title	Metric	Value	Target	Unit	Margin %	Status
:iso_50001	:energy_review	Energy review of the process	:specific_energy	2.01	12.5	:gj_per_t	83.94	Compliant
:iso_50001	:enpi	Energy performance indicator	:spec_energy_vs_design	16.06	100	:pct	83.94	Compliant
:iso_50001	:baseline	Baseline data quality	:completeness	99.81	98	:pct	1.85	Compliant
:iso_50001	:objectives	Objectives and targets	:targets_met	89.29	70	:pct	27.55	Compliant
:iso_50001	:operational_control	Operational control	:critical_findings	30	0	:count	n/a	Noncompliant
:iso_50001	:monitoring	Monitoring and measurement	:metered_signals	69	40	:count	72.5	Compliant
:iso_14001	:aspects	Significant environmental aspects	:stack_fluoride	2.72	5	:count	45.65	Compliant
:iso_14001	:emissions	Emissions monitoring	:stack_dust	21.8	30	:count	27.35	Compliant
:iso_14001	:waste	Waste management	:gypsum_free_p2o5	0.4985	1	:wt_pct	50.15	Compliant
:iso_14001	:water	Water balance and discharge	:water_per_p2o5	3.38	8	:t_ph	57.74	Compliant
:iso_14001	:emergency	Emergency preparedness	:scrubber_ph	6.4	5.6	:count	14.21	Compliant
:iso_46001	:water_review	Water use review	:water_per_p2o5	3.38	8	:t_ph	57.74	Compliant
:iso_46001	:reuse	Water reuse	:water_recovery	70.32	65	:pct	8.19	Compliant
:iso_46001	:effluent	Effluent quality	:effluent_ph	7.4	8.5	:count	12.94	Compliant
:iso_9001	:product_spec	Product specification	:product_n	18.14	17.4	:wt_pct	4.27	Compliant
:iso_9001	:moisture	Product moisture	:product_moisture	1.75	2	:wt_pct	12.39	Compliant
:iso_9001	:solubility	Product solubility	:product_wsp	88.51	85	:wt_pct	4.12	Compliant
:iso_9001	:process_control	Control of production	:acid_strength	55.85	50.5	:wt_pct	10.6	Compliant
:iso_9001	:acid_specification	Merchant acid specification	:acid_so4	1.16	1.5	:wt_pct	22.39	Compliant
:iso_9001	:acid_fluorine	Fluorine of the merchant acid	:acid_f	0.3533	0.5	:wt_pct	29.35	Compliant
:iso_9001	:acid_solids	Solids of the merchant acid	:acid_solids	0.2943	0.5	:wt_pct	41.13	Compliant
:iso_9001	:acid_yield	Yield of the acid route	:acid_p2o5_yield	97.68	96.5	:pct	1.22	Compliant
:ghg_protocol	:scope_1	Scope 1 quantification	:scope_1_t	36,380	48	:t	-75,691	Noncompliant
:ghg_protocol	:scope_2	Scope 2 quantification	:scope_2_t	21,955	240	:t	-9,047.78	Noncompliant
:ghg_protocol	:intensity	Emissions intensity	:carbon_intensity	55.98	145	:kg_ph	61.39	Compliant
:ghg_protocol	:factors	Emission factor register	:factor_count	4	4	:count	0	Compliant
:eu_ied	:bat_emissions	BAT emission levels	:stack_fluoride	2.72	5	:count	45.65	Compliant
:eu_ied	:energy_efficiency	Energy efficiency of BAT	:specific_energy	2.01	12.5	:gj_per_t	83.94	Compliant

Framework	Clause	Title	Metric	Value	Target	Unit	Margin %	Status
:eu_ied	:monitoring	Continuous monitoring	:completeness	99.81	98	:pct	1.85	Compliant
:eu_ied	:waste	Waste management plan	:gypsum_free_p2o5	0.4985	1	:wt_pct	50.15	Compliant
:us_epa_40cfr63	:fluoride	Fluoride emission limit	:stack_fluoride	2.72	5	:count	45.65	Compliant
:us_epa_40cfr63	:scrubber	Scrubber operating limit	:scrubber_ph	6.4	5.6	:count	14.21	Compliant
:us_epa_40cfr63	:records	Record keeping	:completeness	99.81	98	:pct	1.85	Compliant
:ifa_bat	:recovery	P2O5 recovery of the route	:p2o5_recovery	83.65	94	:pct	-11.01	Noncompliant
:ifa_bat	:acid_demand	Specific acid demand	:acid_consumption	2.82	2.85	:t_ph	0.9321	Compliant
:ifa_bat	:acid_steam	Steam of the acid concentration	:acid_steam	0.4115	0.62	:t_ph	33.63	Compliant
:ifa_bat	:water_soluble	Water-soluble loss	:water_soluble_loss	2.32	2	:pct	-15.93	Noncompliant
:ifa_bat	:fluorine	Fluorine recovery	:fluorine_recovery	92	80	:pct	15	Compliant
:ifa_bat	:grind	Liberation of the ore	:grind_p80	150.6	170	:um	11.41	Compliant
:phosphogypsum_code	:stack_quality	Gypsum quality to the stack	:gypsum_free_p2o5	0.4985	1	:wt_pct	50.15	Compliant
:phosphogypsum_code	:water_return	Water return to the plant	:water_recovery	70.32	65	:pct	8.19	Compliant
:phosphogypsum_code	:dust	Dust control on the stack	:stack_dust	21.8	30	:count	27.35	Compliant

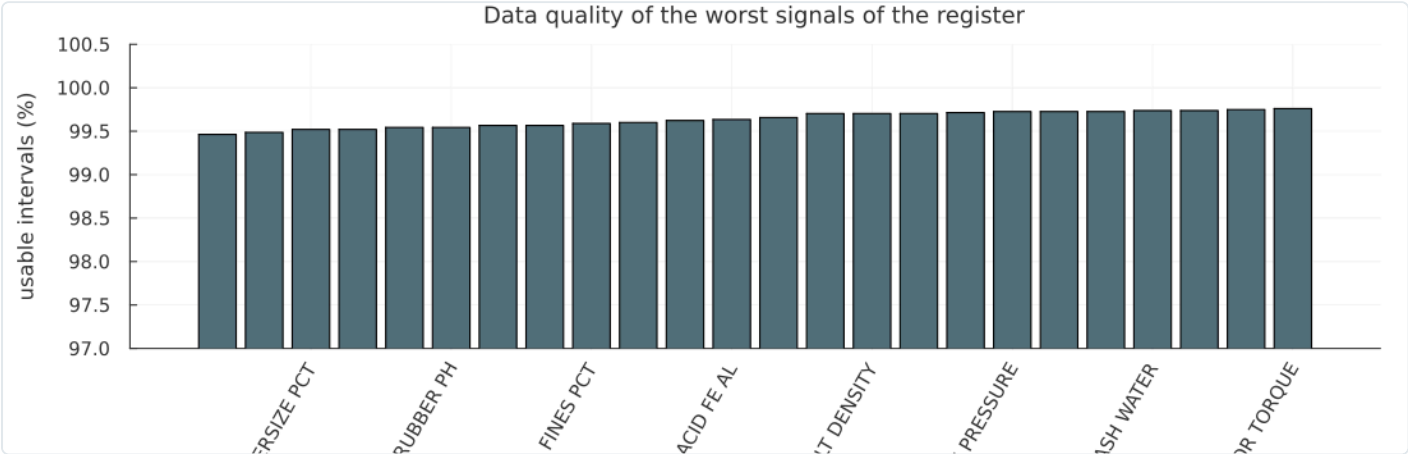
WHAT HAS TO BE DONE ABOUT IT

Framework	Clause	Metric	Value	Target	Gap %	Status	Corrective action
:ghg_protocol	:scope_1	:scope_1_t	36,380	48	-75,691	Noncompliant	Review the operating window with the process engineer
:ghg_protocol	:scope_2	:scope_2_t	21,955	240	-9,047.78	Noncompliant	Review the operating window with the process engineer
:ifa_bat	:water_soluble	:water_soluble_loss	2.32	2	-15.93	Noncompliant	Review the operating window with the process engineer
:ifa_bat	:recovery	:p2o5_recovery	83.65	94	-11.01	Noncompliant	Review the reagent dose and the grind of the blend
:iso_50001	:operational_control	:critical_findings	30	0	n/a	Noncompliant	Close the critical findings of the diagnostics register

Data quality, reconciliation and the instrument register

QUALITY OF THE HISTORIAN

Quantity	Value
Intervals	8,760 hourly intervals of 69 signals
Operating intervals	8,568
Completeness	99.81 % of the register
Defect runs	139 runs over 1151 intervals
Causes	:faulty 568, :measured 602656, :missing 583, :substituted 633
Status	:compliant



Completeness of the historian, the worst signals of the instrument register

EVERY RUN OF UNUSABLE DATA OF THE CAMPAIGN

Signal	From	To	Intervals	Cause
:ACID_COLOUR	2026-01-02T01:00:00	2026-01-02T10:00:00	10	Faulty
:OVERSIZE_PCT	2026-01-02T22:00:00	2026-01-03T11:00:00	14	Missing
:OVERSIZE_PCT	2026-01-09T03:00:00	2026-01-09T10:00:00	8	Missing
:CYCLONE_P80	2026-01-15T16:00:00	2026-01-16T03:00:00	12	Missing
:ACID_SO4	2026-01-21T10:00:00	2026-01-21T22:00:00	13	Faulty
:AMMONIA_FLOW	2026-01-22T08:00:00	2026-01-22T11:00:00	4	Missing
:PROCESS_WATER_FLOW	2026-01-24T01:00:00	2026-01-24T06:00:00	6	Missing
:F_WASH_WATER	2026-01-29T16:00:00	2026-01-29T18:00:00	3	Missing
:PRODUCT_WSP	2026-02-01T03:00:00	2026-02-01T10:00:00	8	Faulty
:PRODUCT_MOISTURE	2026-02-07T13:00:00	2026-02-08T00:00:00	12	Faulty
:DRYER_OUTLET_TEMP	2026-02-07T16:00:00	2026-02-07T22:00:00	7	Missing
:FINES_PCT	2026-02-07T20:00:00	2026-02-07T22:00:00	3	Faulty
:SLURRY_SG	2026-02-14T01:00:00	2026-02-14T08:00:00	8	Missing
:STACK_DUST	2026-02-14T23:00:00	2026-02-15T04:00:00	6	Faulty
:STACK_DUST	2026-02-18T02:00:00	2026-02-18T14:00:00	13	Faulty
:TAIL_P2O5	2026-02-21T11:00:00	2026-02-21T16:00:00	6	Missing
:OVERSIZE_PCT	2026-02-22T23:00:00	2026-02-23T08:00:00	10	Faulty
:DRYER_OUTLET_TEMP	2026-02-24T15:00:00	2026-02-24T21:00:00	7	Faulty
:RECYCLE_RATIO	2026-02-25T06:00:00	2026-02-25T11:00:00	6	Faulty
:COOLER_OUTLET_TEMP	2026-02-28T01:00:00	2026-02-28T04:00:00	4	Missing
:COOLING_SUPPLY_T	2026-03-05T02:00:00	2026-03-05T05:00:00	4	Faulty
:OVERSIZE_PCT	2026-03-05T05:00:00	2026-03-05T11:00:00	7	Faulty

Signal	From	To	Intervals	Cause
:SULPHUR_FEED	2026-03-07T14:00:00	2026-03-08T00:00:00	11	Missing
:PRODUCT_WSP	2026-03-09T17:00:00	2026-03-10T05:00:00	13	Faulty
:COOLER_OUTLET_TEMP	2026-03-09T20:00:00	2026-03-10T00:00:00	5	Missing
:ACID_FE_AL	2026-03-12T09:00:00	2026-03-12T21:00:00	13	Missing
:CONVERTER_TEMP	2026-03-13T08:00:00	2026-03-13T16:00:00	9	Missing
:OVERSIZE_PCT	2026-03-13T18:00:00	2026-03-13T20:00:00	3	Faulty
:ROCK_FEED	2026-03-17T22:00:00	2026-03-18T11:00:00	14	Faulty
:ACID_SOLIDS	2026-03-18T18:00:00	2026-03-18T23:00:00	6	Faulty
:PROCESS_WATER_FLOW	2026-03-19T18:00:00	2026-03-20T03:00:00	10	Faulty
:SCRUBBER_PH	2026-03-21T03:00:00	2026-03-21T10:00:00	8	Missing
:MILL_SOUND_DB	2026-04-01T14:00:00	2026-04-01T20:00:00	7	Faulty
:DRYER_OUTLET_TEMP	2026-04-01T18:00:00	2026-04-02T05:00:00	12	Faulty
:CONVERTER_TEMP	2026-04-03T15:00:00	2026-04-04T04:00:00	14	Missing
:STRONG_ACID_P2O5	2026-04-11T23:00:00	2026-04-12T01:00:00	3	Faulty
:CYCLONE_PRESSURE	2026-04-13T09:00:00	2026-04-13T18:00:00	10	Missing
:CONC_P2O5	2026-04-14T15:00:00	2026-04-14T17:00:00	3	Missing
:MILL_POWER	2026-04-16T18:00:00	2026-04-16T21:00:00	4	Missing
:TAIL_P2O5	2026-04-18T14:00:00	2026-04-19T01:00:00	12	Faulty
:GRANULATOR_TORQUE	2026-04-24T01:00:00	2026-04-24T03:00:00	3	Missing
:FILTER_VACUUM	2026-04-29T17:00:00	2026-04-29T21:00:00	5	Faulty
:WASH_WATER_FLOW	2026-04-29T21:00:00	2026-04-30T04:00:00	8	Missing
:ACID_F	2026-05-01T23:00:00	2026-05-02T01:00:00	3	Faulty
:ACID_FE_AL	2026-05-03T04:00:00	2026-05-03T11:00:00	8	Missing
:ACID_F	2026-05-04T03:00:00	2026-05-04T15:00:00	13	Missing
:CYCLONE_P80	2026-05-05T06:00:00	2026-05-05T12:00:00	7	Missing
:PRODUCT_FLOW	2026-05-08T07:00:00	2026-05-08T15:00:00	9	Faulty
:CYCLONE_P80	2026-05-11T12:00:00	2026-05-11T20:00:00	9	Faulty
:SCRUBBER_PH	2026-05-17T02:00:00	2026-05-17T13:00:00	12	Faulty
:CONVERTER_TEMP	2026-05-18T10:00:00	2026-05-18T19:00:00	10	Missing
:SCRUBBER_PH	2026-05-20T09:00:00	2026-05-20T22:00:00	14	Missing
:WEAK_ACID_P2O5	2026-05-21T04:00:00	2026-05-21T13:00:00	10	Faulty
:CRUSHER_POWER	2026-05-24T17:00:00	2026-05-25T01:00:00	9	Faulty
:GYPSUM_FLOW	2026-06-01T03:00:00	2026-06-01T05:00:00	3	Faulty
:TURBINE_POWER	2026-06-01T21:00:00	2026-06-02T03:00:00	7	Missing
:FINES_PCT	2026-06-02T06:00:00	2026-06-02T09:00:00	4	Missing
:TURBINE_POWER	2026-06-04T14:00:00	2026-06-05T00:00:00	11	Missing
:GRAN_BED_TEMP	2026-06-04T16:00:00	2026-06-04T18:00:00	3	Faulty
:FLOT_FEED_P2O5	2026-06-05T21:00:00	2026-06-06T01:00:00	5	Faulty
:GYPSUM_FREE_P2O5	2026-06-10T07:00:00	2026-06-10T20:00:00	14	Missing
:STEAM_FLOW	2026-06-17T08:00:00	2026-06-17T10:00:00	3	Missing
:GYPSUM_FLOW	2026-06-21T06:00:00	2026-06-21T10:00:00	5	Faulty
:MELT_DENSITY	2026-06-22T21:00:00	2026-06-23T08:00:00	12	Faulty
:CYCLONE_P80	2026-06-23T13:00:00	2026-06-24T02:00:00	14	Missing
:MELT_DENSITY	2026-06-29T08:00:00	2026-06-29T11:00:00	4	Missing
:EVAP_FEED_FLOW	2026-06-29T16:00:00	2026-06-29T18:00:00	3	Faulty

Signal	From	To	Intervals	Cause
:RECYCLE_ACID_FLOW	2026-07-02T11:00:00	2026-07-02T21:00:00	11	Faulty
:GRANULATOR_TORQUE	2026-07-06T23:00:00	2026-07-07T02:00:00	4	Missing
:F_WASH_WATER	2026-07-10T05:00:00	2026-07-10T14:00:00	10	Missing
:SULPHUR_FEED	2026-07-10T20:00:00	2026-07-11T08:00:00	13	Missing
:ACID_COLOUR	2026-07-15T11:00:00	2026-07-15T21:00:00	11	Missing
:ROM_FEED	2026-07-15T22:00:00	2026-07-16T10:00:00	13	Faulty
:FILTER_VACUUM	2026-07-17T03:00:00	2026-07-17T10:00:00	8	Faulty
:CYCLONE_PRESSURE	2026-07-17T10:00:00	2026-07-17T18:00:00	9	Faulty
:WASH_WATER_FLOW	2026-07-18T08:00:00	2026-07-18T18:00:00	11	Faulty
:MERCHANT_ACID_FLOW	2026-07-22T17:00:00	2026-07-23T06:00:00	14	Faulty
:CONC_P2O5	2026-07-27T14:00:00	2026-07-28T02:00:00	13	Missing
:RECYCLE_RATIO	2026-07-28T20:00:00	2026-07-29T03:00:00	8	Missing
:PRODUCT_FLOW	2026-07-30T20:00:00	2026-07-30T23:00:00	4	Faulty
:STRONG_ACID_P2O5	2026-07-30T23:00:00	2026-07-31T05:00:00	7	Missing
:F_WASH_WATER	2026-07-31T00:00:00	2026-07-31T09:00:00	10	Missing
:GRANULATOR_TORQUE	2026-08-01T13:00:00	2026-08-01T19:00:00	7	Missing
:PRODUCT_MOISTURE	2026-08-01T16:00:00	2026-08-02T02:00:00	11	Missing
:ROCK_P2O5	2026-08-04T14:00:00	2026-08-04T20:00:00	7	Faulty
:SLURRY_SG	2026-08-07T20:00:00	2026-08-07T22:00:00	3	Missing
:EVAP_VACUUM	2026-08-08T13:00:00	2026-08-09T01:00:00	13	Faulty
:PRODUCT_FLOW	2026-08-20T16:00:00	2026-08-21T03:00:00	12	Missing
:CYCLONE_P80	2026-08-22T02:00:00	2026-08-22T04:00:00	3	Faulty
:ROCK_FEED	2026-08-27T03:00:00	2026-08-27T05:00:00	3	Faulty
:RECYCLE_RATIO	2026-09-04T03:00:00	2026-09-04T08:00:00	6	Faulty
:FREE_SO4	2026-09-04T05:00:00	2026-09-04T10:00:00	6	Missing
:PRODUCT_FLOW	2026-09-06T22:00:00	2026-09-07T03:00:00	6	Missing
:COOLING_RETURN_T	2026-09-08T05:00:00	2026-09-08T14:00:00	10	Missing
:FLOT_FEED_P2O5	2026-09-10T03:00:00	2026-09-10T16:00:00	14	Faulty
:FLOT_FEED_P2O5	2026-09-12T07:00:00	2026-09-12T14:00:00	8	Faulty
:SCRUBBER_PH	2026-09-12T20:00:00	2026-09-13T01:00:00	6	Faulty
:STEAM_FLOW	2026-09-13T12:00:00	2026-09-13T22:00:00	11	Missing
:H2SO4_FLOW	2026-09-13T15:00:00	2026-09-14T03:00:00	13	Missing
:MELT_DENSITY	2026-09-13T17:00:00	2026-09-14T02:00:00	10	Missing
:GRAN_BED_TEMP	2026-09-17T09:00:00	2026-09-17T11:00:00	3	Missing
:FLOT_FEED_P2O5	2026-09-20T18:00:00	2026-09-21T06:00:00	13	Missing
:WASH_WATER_FLOW	2026-09-24T12:00:00	2026-09-24T18:00:00	7	Faulty
:EVAP_DENSITY	2026-10-03T04:00:00	2026-10-03T07:00:00	4	Faulty
:FILTER_RATE	2026-10-04T04:00:00	2026-10-04T11:00:00	8	Faulty
:CYCLONE_PRESSURE	2026-10-04T05:00:00	2026-10-04T09:00:00	5	Missing
:MERCHANT_ACID_FLOW	2026-10-07T11:00:00	2026-10-07T22:00:00	12	Faulty
:ROCK_P2O5	2026-10-13T20:00:00	2026-10-14T09:00:00	14	Missing
:FINES_PCT	2026-10-18T17:00:00	2026-10-19T00:00:00	8	Faulty
:WASH_WATER_FLOW	2026-10-18T21:00:00	2026-10-19T05:00:00	9	Faulty
:GRANULATOR_TORQUE	2026-10-20T19:00:00	2026-10-21T01:00:00	7	Faulty
:FILTER_VACUUM	2026-10-23T00:00:00	2026-10-23T05:00:00	6	Faulty

Signal	From	To	Intervals	Cause
:ACID_COLOUR	2026-11-02T00:00:00	2026-11-02T03:00:00	4	Faulty
:FILTER_VACUUM	2026-11-07T08:00:00	2026-11-07T10:00:00	3	Missing
:PRODUCT_WSP	2026-11-12T14:00:00	2026-11-13T02:00:00	13	Faulty
:ACID_SOLIDS	2026-11-14T01:00:00	2026-11-14T14:00:00	14	Missing
:SO2_CONC	2026-11-15T08:00:00	2026-11-15T16:00:00	9	Missing
:FINES_PCT	2026-11-15T22:00:00	2026-11-16T05:00:00	8	Missing
:PRODUCT_WSP	2026-11-18T07:00:00	2026-11-18T14:00:00	8	Missing
:FINES_PCT	2026-11-18T09:00:00	2026-11-18T21:00:00	13	Faulty
:CRUSHER_POWER	2026-11-21T06:00:00	2026-11-21T12:00:00	7	Missing
:SULPHUR_FEED	2026-11-22T08:00:00	2026-11-22T21:00:00	14	Missing
:MILL_FEED_WATER	2026-11-22T12:00:00	2026-11-22T17:00:00	6	Missing
:MILL_FEED_WATER	2026-11-23T12:00:00	2026-11-23T14:00:00	3	Missing
:FILTER_RATE	2026-11-23T14:00:00	2026-11-23T17:00:00	4	Missing
:FILTER_RATE	2026-11-24T00:00:00	2026-11-24T11:00:00	12	Faulty
:GRAN_BED_TEMP	2026-11-24T09:00:00	2026-11-24T12:00:00	4	Missing
:PRODUCT_FLOW	2026-11-28T22:00:00	2026-11-29T04:00:00	7	Faulty
:GYPSUM_FLOW	2026-12-06T15:00:00	2026-12-06T22:00:00	8	Faulty
:RECYCLE_ACID_FLOW	2026-12-07T01:00:00	2026-12-07T09:00:00	9	Faulty
:ROCK_FEED	2026-12-13T17:00:00	2026-12-13T20:00:00	4	Faulty
:GRAN_BED_TEMP	2026-12-13T21:00:00	2026-12-14T10:00:00	14	Missing
:TURBINE_POWER	2026-12-19T05:00:00	2026-12-19T12:00:00	8	Missing
:ROCK_FEED	2026-12-22T09:00:00	2026-12-22T22:00:00	14	Faulty
:ACID_FE_AL	2026-12-22T10:00:00	2026-12-22T20:00:00	11	Faulty
:WASH_WATER_FLOW	2026-12-24T18:00:00	2026-12-25T05:00:00	12	Missing
:GYPSUM_FREE_P2O5	2026-12-25T13:00:00	2026-12-25T17:00:00	5	Faulty
:MERCHANT_ACID_FLOW	2026-12-27T04:00:00	2026-12-27T07:00:00	4	Faulty
:EVAP_DENSITY	2026-12-30T22:00:00	2026-12-31T03:00:00	6	Faulty

The flows of the campaign were reconciled against the stoichiometry of the reaction section: the weighted least squares adjustment leaves a global test statistic of 21,779 against a critical value of 11.07 for 5 balances, so the measurements are inconsistent with the balances.

Signal	Measured	Reconciled	Adjustment	In sigma	Unit	Gross error
:P2O5_FEED_RATE	121.62	38.16	-83.46	-104.32	:t_ph	1
:H2SO4_FLOW	343.39	377.22	33.83	11.28	:t_ph	1
:SULPHUR_FEED	112.22	217.76	105.54	87.95	:t_ph	1
:GYPSUM_FLOW	565.63	764.55	198.93	49.73	:t_ph	1
:AMMONIA_FLOW	44.3	58.28	13.98	17.47	:t_ph	1
:PRODUCT_FLOW	199.92	263.71	63.78	15.95	:t_ph	1

Method, units and assumptions

Area	Streams	Equipment	Unit operations	Instrument tags
Beneficiation		8	9	4
Flotation		9	5	4
Attack		5	3	3
Filtration		6	5	4
Evaporation		3	3	3
SulphuricAcidPlant		6	7	4
Granulation		15	8	8
Utilities		8	3	2
Tailings		1	1	2

- The **design** of the plant is registered in ``process.jl``: capacities, targets, the flowsheet and the instrument register. Every model of the package reads it, so a capacity change propagates to the twin, the optimiser, the controller and the report.
- The **campaign** is a seeded grey-box model of that design (``plantdata.jl``): the production schedule, the ore blend, the weather and the injected deviations are all functions of the seed, so every figure and every PDF can be reproduced from the seed printed on the cover page.
- The **stoichiometry** is computed from the molar masses in ``measures.jl``: 2.3 t of H2SO4 and 4.04 t of gypsum per tonne of P2O5 in fluorapatite, plus the demand of the free carbonate of the ore. The assays of the ore bodies are representative published figures, not the assay of a particular mine.
- The **twin** is written in ModelingToolkit: a square nonlinear balance of 15 unknowns for the steady state, and 11 differential equations on an hourly clock for the dynamics. Both report the residual of every equation they solve.
- The **controller** is a linear MPC over the linearised twin with a Kalman filter, closed on the nonlinear twin as the plant; the baseline is a PI controller tuned on the same steady-state gains, so the comparison is like for like.
- The **soft sensors** are linear models of the variables that are measured slowly or not at all, validated on a holdout and equipped with an applicability domain: outside the conditions they were fitted on, they report that they cannot answer.
- The **cost and carbon** of the site use the register of ``process.jl`` (USD) and the factors of the GHG Protocol: 0.371 kgCO2e/kWh for the grid and 50.29 kgCO2e/GJ for the fuel.

Generated by PHOSPHATE from the baseline and the seed recorded on the cover page.